

Iwasawa Theory for Elliptic Curves

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Chapter 1

Introduction

In 1959, Kenkichi Iwasawa (岩澤健吉) published an article [1] in which he studied Galois $\mathbf{Z}_p$ -extensions. For a number field F , one can build an extension F_∞/F whose Galois group is isomorphic to $\mathbf{Z}_p$. Then, by Galois theory, we can associate a subextension F_n/F of degree p^n for each subgroup $p^n\mathbf{Z}_p$ where n runs through $\mathbf{N}$. Surprisingly, even though the study of the intermediate finite extensions F_n/F was considered as a hard challenge, Iwasawa showed that the infinite extension F_∞/F could shed a light on its finite subextensions. More precisely, he showed that the exact p -power p^{e_n} of the class number of F_n/F satisfied $e_n = \mu p^n + \lambda n + \nu$ where μ , λ and ν are constant terms. This is the content of the celebrated Iwasawa theorem.

The ideas from Iwasawa theory inspired Mazur in the study of elliptic curves. Instead of considering the class numbers of the intermediate extensions, Mazur was interested in the study of the p -primary Selmer group $\text{Sel}_E(F_n)_p$ of an elliptic curve E defined over F . Using methods of Iwasawa theory, Mazur proved that $\text{Sel}_E(F_\infty)_p$ controls the $\text{Sel}_E(F_n)$'s in 1972 [2].

These Selmer groups were introduced by Selmer in 1954 [3] to study the failure of a local-global principle for equations of degree higher than 2. In 1959, Cassels [4] formalized the modern cohomological definition of these groups. Selmer groups play a pivotal role in number theory, being a central object of the Birch and Swinnerton-Dyer conjecture [5, 6, 7] that establishes a deep link between analysis and algebra: it predicts that the vanishing order at 1 of the L -function associated to E is equal to the rank of the finitely generated abelian group $E(F)$.

In this report, we present the work of Greenberg [8] on the Iwasawa theory for elliptic curves. Greenberg formalized Mazur's control theorem in a cohomological setting.

Chapter 2

Algebraic preliminaries

2.1 Galois cohomology

In this section, we recall results from group cohomology and Galois cohomology. We refer the reader to Brown [9] and Serre [10] for an introduction to these topics.

Let G be a topological group (when the topology is not precised, we always assume the topology to be discrete). Let M be a topological $\mathbf{Z}G$ -module ($\mathbf{Z}G$ denotes the free abelian group generated by G).

Definition 2.1.1. Let $n \in \mathbf{N}$. We denote by $C^n(G, M)$ the set of continuous functions from G^n to M . For any $\varphi \in C^n(G, M)$ let $d\varphi \in C^{n+1}(G, M)$ be given by

$$\begin{aligned} d(\varphi)(g_0, \dots, g_n) &= g_0\varphi(g_1, \dots, g_n) - \varphi(g_0g_1, g_2, \dots, g_n) + \varphi(g_0, g_1g_2, \dots, g_n) \\ &\quad + \dots + (-1)^{n-1}\varphi(g_0, \dots, g_{n-1}g_n) + (-1)^n\varphi(g_0, \dots, g_{n-1}). \end{aligned}$$

One can check that $d^2 = 0$. Hence, this gives a complex chain

$$M \simeq C^0(G, M) \xrightarrow{d} C^1(G, M) \xrightarrow{d} \dots C^n(G, M) \xrightarrow{d} \dots$$

As usual in cohomology, we denote by $H^n(G, M)$ the n -th cohomology group associated to $C^*(G, M)$.

Let $m \in M$. Then, we have

$$\forall g \in G, d(m)(g) = gm - m.$$

Hence, $dm = 0$ (in other words, m is a 0-cocycle) if and only if $gm = m$. Therefore, $H^0(G, M) \cong M^G$.

Now, let $\varphi \in C^1(G, M)$. We have

$$\forall (g_0, g_1) \in G^2, d\varphi(g_0, g_1) = g_0\varphi(g_1) - \varphi(g_0g_1) + \varphi(g_0).$$

Hence, $d\varphi = 0$ if and only if

$$\forall (g_0, g_1) \in G^2, \varphi(g_0 g_1) = g_0 \varphi(g_1) + \varphi(g_0).$$

We call such a φ a *crossed morphism*.

Consider an exact sequence of

$$1 \longrightarrow M \longrightarrow E \longrightarrow G \longrightarrow 1$$

where E is a group. This sequence splits if and only if $H^2(G, M) = 0$. This is the case in particular when M is divisible (see Serre [11, Chapter XIII, Proposition 2]).

When G is cyclic, its cohomology groups are periodic.

Proposition 2.1.2. *Suppose G cyclic generated by σ . Then, for any integer q , we have*

$$H^{2q+2}(G, M) = \frac{\ker(\sigma - 1)}{\text{Im}(\sum_{\tau \in G} \tau)}$$

and

$$H^{2q+1}(G, M) = \frac{\ker(\sum_{\tau \in G} \tau)}{\text{Im}(\sigma - 1)}.$$

For a proof of this fact, we refer the reader to Serre [10, Chapter VIII, Corollary of Proposition 6].

Now, suppose M is the direct limit of $\varinjlim_n M_n$ taken over a set of morphisms $\{f_{n,m}: M_n \rightarrow M_m, (n, m) \in \mathbf{N}^2, n > m\}$ where the M_n 's are finite $\mathbf{Z}G$ -modules.

Proposition 2.1.3. *The cohomology is compatible with the direct limit:*

$$\forall i \in \mathbf{N}, \varinjlim_n H^i(G, M_n) = H^i(G, M)$$

where the direct limit is taken over the f_{n,m_*} 's (left composition by $f_{n,m}$).

Proof. Let i, m and n be integers such that $m < n$. We need to check the commutativity of the following diagram:

$$\begin{array}{ccc} H^i(G, M_n) & & \\ f_{n,m_*} \downarrow & \searrow f_{n*} & \\ H^i(G, M_m) & \xrightarrow{f_{m*}} & H^i(G, M) \end{array}$$

inside which $f_n: M_n \rightarrow M$ and $f_m: M_m \rightarrow M$ send any element of M_n and M_m respectively to its equivalence class in M . The commutativity follows from the equality $f_m \circ f_{n,m} = f_n$. $\square$

From a short exact sequence, one can build a *Galois cohomology long exact sequence*. Indeed, from a short exact sequence of G -modules M, M' and M''

$$0 \longrightarrow M \longrightarrow M' \longrightarrow M'' \longrightarrow 0$$

we can build an exact sequence

$$\begin{array}{ccccccc}
0 & \longrightarrow & H^0(G, M) & \longrightarrow & H^0(G, M') & \longrightarrow & H^0(G, M'') & \longrightarrow & \cdots \\
& & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
& & H^1(G, M) & \longrightarrow & H^1(G, M') & \longrightarrow & H^1(G, M'') & \longrightarrow & \cdots \\
& & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
& & \cdots & & \cdots & & \cdots & & \cdots \\
& & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
& & H^n(G, M) & \longrightarrow & H^n(G, M') & \longrightarrow & H^n(G, M'') & \longrightarrow & \cdots \\
& & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
& & \cdots & & \cdots & & \cdots & & \cdots
\end{array}$$

Now, consider a normal subgroup $N \triangleleft G$. One can build the *inflation-restriction sequence* which is the following exact sequence.

$$0 \rightarrow H^1(G/N, M^N) \xrightarrow{\text{inf}} H^1(G, M) \xrightarrow{\text{res}} H^1(N, M)^{G/N} \xrightarrow{\text{tr}} H^2(G/N, M^N)$$

where inf, res and tr are respectively the *inflation*, *restriction* and *transgression* morphisms. Note that $H^1(N, M)^{G/N}$ denotes the elements of $H^1(N, M)$ fixed by the action of G/N . The action of $g \in G$ on a cocycle φ is given by

$$\forall n \in N, (g \cdot \varphi)(n) = g \cdot \varphi(g^{-1}ng).$$

This depends only on the class of g modulo N and on the cohomology class of φ . Hence, this gives the action of G/N on $H^1(N, M)$.

Let F be a number field and m be an integer. Let ζ be a primitive p^m -root of unity. The Galois group $\text{Gal}(\overline{F}/F)$ acts on p^m -roots of unity. In particular, we have $\zeta^\sigma = \zeta^{n_\sigma}$ for some integer n_σ invertible modulo p^m . Then, we can define $\chi_m: \text{Gal}(\overline{F}/F) \rightarrow (\mathbf{Z}/p^m\mathbf{Z})^\times$ by $\chi_m(\sigma) = n_\sigma$. This endows $\mathbf{Z}/p^m\mathbf{Z}$ with a structure of $\text{Gal}(\overline{F}/F)$ -module. Now, notice that $\chi_{m+1} \equiv \chi_m \pmod{p^m}$. Hence, the projective limit inherits the module structure.

Definition 2.1.4. We denote by $\mathbf{Z}_p(1)$ the projective limit of the $\mathbf{Z}/p^n\mathbf{Z}$'s endowed with the $\text{Gal}(\overline{F}/F)$ -module structure inherited by the χ_n 's. We call *cyclotomic character* the representation $\chi: \text{Gal}(\overline{F}/F) \rightarrow \mathbf{Z}_p^\times$.

More generally, for a p -adic local field k and a $\text{Gal}(\overline{k}/k)$ -module M whose action is given by ψ , we denote by $M(1)$ the group M endowed with the action $\chi\psi$. The following celebrated theorem of Poitou-Tate duality establishes isomorphisms between low-dimensional cohomology groups of M and $M(1)$ (see Neukirch, Schmidt and Wingberg [12, Theorem 7.2.9]).

Theorem 2.1.5 (Poitou-Tate duality). *Let k be a p -adic local field of characteristic zero and M be a finitely generated $\text{Gal}(\overline{k}/k)$ -module. Then, for any $i \in \{0, 1, 2\}$, there exists a canonical isomorphism*

$$H^i(k, M(1)) \longleftrightarrow H^{2-i}(k, M)^*.$$

Another fruitful theorem is Hilbert's Theorem 90 that establishes the vanishing of the first Galois cohomology group of the multiplicative group of a field (see Neukirch, Schmidt and Wingberg [12, Theorem 6.2.1]).

Theorem 2.1.6 (Hilbert's Satz 90). *Let L/K be a Galois extension. Then*

$$H^1(\text{Gal}(L/K), L^\times) = 0.$$

2.2 Pontryagin duality

In this section, we recall some results about Pontryagin duality. For a detailed account on Pontryagin duality, see Folland [13].

For any integer m and for any abelian group G , let us denote by $G[m]$ the kernel of $[m]$, i.e., the m -torsion points of G . Moreover, for any abelian group G , let us denote by $G[p^\infty]$ the set $\bigcup_{m \in \mathbb{N}} G[p^m]$ for any prime p , i.e., the p -power torsion subgroup of G , and by G_{tors} the set $\bigcup_{m \in \mathbb{N}} G[m]$, i.e., the torsion subgroup of G .

In this section, we fix A , an abelian group endowed with a discrete topology.

Definition 2.2.1 (Pontryagin dual). The Pontryagin dual of A is

$$\widehat{A} = \text{Hom}_{\text{cont}}(A, \mathbf{R}/\mathbf{Z}).$$

From now on, assume that A is p -primary, i.e., $A[p^\infty] = A$, for some prime number p .

Proposition 2.2.2. *The Pontryagin dual of A is*

$$\widehat{A} = \text{Hom}(A, \mathbf{Q}_p/\mathbf{Z}_p).$$

Proof. Let $\varphi \in \widehat{A}$. Since A is p -primary, $\text{Im}(\varphi)$ is also p -primary. Hence, φ is a continuous morphism from A to $\mathbf{R}/\mathbf{Z}[p^\infty]$ which is isomorphic to $\mathbf{Q}_p/\mathbf{Z}_p$. This last group is discrete so any morphism from A to $\mathbf{Q}_p/\mathbf{Z}_p$ is continuous. $\square$

Definition 2.2.3. The $\mathbf{Z}_p$ -corank of A , denoted by $\text{corank}_{\mathbf{Z}_p}(A)$ is the $\mathbf{Z}_p$ -rank of $\widehat{A}$.

Proposition 2.2.4. *Let $m \in \mathbb{N}$. The Pontryagin dual of $(\mathbf{Q}_p/\mathbf{Z}_p)^m$ is $\mathbf{Z}_p^m$.*

Proof. Let $m \in \mathbb{N}$. By definition, we have

$$\begin{aligned} (\widehat{\mathbf{Q}_p/\mathbf{Z}_p})^m &= \text{Hom}((\mathbf{Q}_p/\mathbf{Z}_p)^m, \mathbf{Q}_p/\mathbf{Z}_p) = \text{Hom}(\varprojlim_n (\mathbf{Z}/p^n \mathbf{Z})^m, \mathbf{Q}_p/\mathbf{Z}_p) \\ &= \varprojlim_n \text{Hom}((\mathbf{Z}/p^n \mathbf{Z})^m, \mathbf{Q}_p/\mathbf{Z}_p) = \varprojlim_n \text{Hom}(\mathbf{Z}/p^n \mathbf{Z}, \mathbf{Q}_p/\mathbf{Z}_p)^m. \end{aligned}$$

Then, for any integer n , the set $\text{Hom}(\mathbf{Z}/p^n \mathbf{Z}, \mathbf{Q}_p/\mathbf{Z}_p)$ is isomorphic to $\mathbf{Z}/p^n \mathbf{Z}$ because a morphism from $\text{Hom}(\mathbf{Z}/p^n \mathbf{Z}, \mathbf{Q}_p/\mathbf{Z}_p)$ is uniquely defined by its image of 1 (which should have order at most p^n). Hence, the result follows from the equality $\mathbf{Z}_p^m = \varprojlim_n (\mathbf{Z}/p^n \mathbf{Z})^m$. $\square$

Definition 2.2.5. The group A is divisible if the multiplication by n is a surjective map for any non-zero integer n .

Proposition 2.2.6. *If A is divisible of finite $\mathbf{Z}_p$ -corank then it is isomorphic to $(\mathbf{Q}_p/\mathbf{Z}_p)^{\text{corank}_{\mathbf{Z}_p}(A)}$ and its Pontryagin dual is $\mathbf{Z}_p^{\text{corank}_{\mathbf{Z}_p}(A)}$.*

Proof. Suppose A divisible and let us denote by r its $\mathbf{Z}_p$ -corank. Since A is p -primary, it can be written as $A = \varinjlim_n A[p^n]$. Moreover, for any integer n , $A[p^n]$ can be written as $(\mathbf{Z}/p^n\mathbf{Z})^{\mathfrak{a}_n}$ where the $\mathfrak{a}_n$'s are cardinals. By divisibility, the multiplication by p is surjective from $A[p^{n+1}]$ to $A[p^n]$ for any integer n . In particular, all the $\mathfrak{a}_n$'s are equal to a cardinal $\mathfrak{a}$. Therefore, A is the inductive limit of the $(\mathbf{Z}/p^n\mathbf{Z})^{\mathfrak{a}}$'s, i.e., $A = (\mathbf{Q}_p/\mathbf{Z}_p)^{\mathfrak{a}}$. The second part of the proposition follows from Proposition 2.2.4. $\square$

Corollary 2.2.7. *Let r be an integer. The divisible subgroups of $(\mathbf{Q}_p/\mathbf{Z}_p)^r$ are the $(\mathbf{Q}_p/\mathbf{Z}_p)^s$ for $s \in \{0, \dots, r\}$.*

Proof. The case $r = 1$ is a direct consequence of Proposition 2.2.6. Now, let r be a positive integer and let H be a divisible subgroup of $(\mathbf{Q}_p/\mathbf{Z}_p)^r$. By the initial case, the image of the r -th projection π is either 0 or $\mathbf{Q}_p/\mathbf{Z}_p$. Thus, the result follows inductively, noticing when $\text{Im}(\pi)$ is not trivial that the following exact sequence splits thanks to the divisibility of $H \cap \ker(\pi)$:

$$0 \longrightarrow H \cap \ker(\pi) \longrightarrow H \xrightarrow{\pi} \mathbf{Q}_p/\mathbf{Z}_p \longrightarrow 0$$

$\square$

Proposition 2.2.8. *Let B be a finite subgroup of A . Then, we have $\widehat{A} \cong \widehat{A/B}$.*

Proof. Let B be a finite subgroup of A . Applying $\text{Hom}(\cdot, \mathbf{Q}_p/\mathbf{Z}_p)$ to the exact sequence

$$0 \longrightarrow B \longrightarrow A \longrightarrow A/B \longrightarrow 0$$

we get the exact sequence

$$0 \longrightarrow \widehat{A/B} \longrightarrow \widehat{A} \longrightarrow \widehat{B}$$

Let φ be a morphism from $\widehat{B}$. Then, its image is a subgroup of $\mathbf{Q}_p/\mathbf{Z}_p$. Hence, it is a finite divisible group. So it is trivial. Thus, $\widehat{B} = 0$ and the result follows from the exact sequence. $\square$

2.3 Iwasawa theory

In this section, we recall the setting of Iwasawa theory. We refer the reader to the book of Washington [14] for an introduction to the subject.

Consider F_∞ , a $\mathbf{Z}_p$ -extension of F , i.e, such that $\text{Gal}(F_\infty/F)$ is isomorphic to $\mathbf{Z}_p$. By Galois theory, we can find subextensions

$$F_\infty \supset \cdots \supset F_n \supset \cdots \supset F_1 \supset F_0 = F$$

such that $\text{Gal}(F_n/F) = \mathbf{Z}/p^n\mathbf{Z}$ for any integer n . In particular, $F_\infty = \bigcup_{n \in \mathbf{N}} F_n$ and $\text{Gal}(F_\infty/F)$ is the projective limits of the $\text{Gal}(F_n/F)$'s taken over the restriction maps.

The most natural example is the following:

Example 2.3.1. Suppose $F = \mathbf{Q}$. Let m be a positive integer. Consider $K_m = \mathbf{Q}(\zeta_{p^m})$ the cyclotomic extension of $\mathbf{Q}$ generated by a primitive p^m -root of unity. The Galois group $\text{Gal}(K_m/\mathbf{Q})$ is isomorphic to $(\mathbf{Z}/p^m\mathbf{Z})^\times$, itself isomorphic to $\Delta \times \mathbf{Z}/p^{m-1}\mathbf{Z}$ where $\Delta = \mathbf{Z}/(p-1)\mathbf{Z}$. We define F_{m-1} as the field K_m^Δ . Then, by Galois theory, $F_{m-1}/\mathbf{Q}$ is a field extension of degree p^{m-1} . One can see that F_m is an extension of F_{m-1} by considering the restriction morphism from $\text{Gal}(K_{m+1}/\mathbf{Q})$ to $\text{Gal}(K_m/\mathbf{Q})$ and noticing that it sends $\mathbf{Z}/p^m\mathbf{Z}$ to $\mathbf{Z}/p^{m-1}\mathbf{Z}$.

More generally, an extension L/F is a pro- p extension if its Galois group is pro- p , i.e., if it is the projective limit (over restriction maps) of p -primary Galois group.

Let S be a set. Let $F(S)$ denote the free discrete group generated by S . Then, the free pro- p group generated by S is the projective limit $\varprojlim F(S)/N$ where N varies through cofinite normal subgroups of $F(S)$ of finite index a power of p . It is known that $H^2(G, M)$ is trivial for any G -module M whenever G is a free pro- p group (see Serre [10, Chapter 1, Section 3, Corollary of Proposition 16]). In particular, this result applies to $\mathbf{Z}_p$, the free pro- p group generated by one element.

Moreover, according to Neukirch, Schmidt and Wingberg [12, Proposition 1.7.7], we have:

Proposition 2.3.2. *Suppose G is a pro- p group topologically generated by σ . Then, we have*

$$H^1(G, M) = \frac{M}{(\sigma - 1)M}.$$

Definition 2.3.3 (group ring). Let R be a ring and G be an abelian group on which R acts. Then, we define the group ring $R[G]$ as the free R -module generated by G endowed with the multiplication map $*$ defined by:

$$\sum_{g \in G} a_g g * \sum_{g \in G} b_g g = \sum_{g \in G} \left(\sum_{hk=g} a_h b_k \right) g$$

for any elements $\sum_{g \in G} a_g g$ and $\sum_{g \in G} b_g g$ of $R[G]$, written additively.

Remark. In the definition above, finitely many a_g and b_g 's are non-zero.

Definition 2.3.4 (completed group ring). Let R be a ring and G be an abelian pro- p group on which R acts, with an action compatible with projection maps. Then, writing G as the projective limit of G_n for $n \in \mathbf{N}$, we define the completed group ring $R[[G]]$ as the projective limit of $R[G_n]$ induced by the projection maps.

In particular, the *Iwasawa algebra* Λ is defined as $\mathbf{Z}_p[[\Gamma]]$ where Γ denotes $\text{Gal}(F_\infty/F)$.

Proposition 2.3.5. *Let γ be a topological generator of Γ , i.e., an element that generates a group dense in Γ . Then, the following map is an isomorphism*

$$\begin{array}{ccc} \mathbf{Z}_p[[T]] & \longrightarrow & \Lambda \\ T & \longmapsto & \gamma - 1 \end{array}.$$

Finally, let us give the structure of finitely generated Λ -modules. First, we need to introduce distinguished polynomials.

Definition 2.3.6. A polynomial $P(T) \in \mathbf{Z}_p[T]$ is *distinguished* if it is of the form

$$P(T) = T^n + a_{n-1}T^{n-1} + \cdots + a_0$$

where n is the degree of P and the a_i 's belong to (p) .

Thanks to Proposition 2.3.5, such polynomials can be seen as elements of Λ . We finish by stating the structure theorem (see Washington [14, Theorem 13.12]).

Theorem 2.3.7 (structure theorem). *Let M be a finitely generated Λ -module. Then, M is pseudo-isomorphic to (in other words, there is a morphism of finite kernel and cokernel from M to)*

$$\Lambda^r \oplus \left(\bigoplus_{i=1}^s \frac{\Lambda}{(p^{n_i})} \right) \oplus \left(\bigoplus_{j=1}^t \frac{\Lambda}{(f_j(T)^{m_j})} \right)$$

where r, s, t , the n_i 's and the m_j 's are integers and the f_j 's are distinguished irreducible polynomials.

Chapter 3

Mazur's control theorem

3.1 Elliptic curves

In this section, we recall basic properties of elliptic curves. Most of the tools we are using can be found in the book of Silverman [15].

Definition 3.1.1 (Elliptic curve). An elliptic curve E defined over a field F is a projective smooth algebraic curve of genus 1 defined over F and that contains an F -point O . For any field extension M/F , we denote by $E(M)$ the set of M -points of E , i.e., the solutions in M of the polynomial equation defining E .

Throughout this report, we fix an elliptic curve E defined over a number field F , i.e., a finite extension of $\mathbf{Q}$. We recall that the elliptic curve E is endowed with an abelian group law whose neutral is O . Hence, it is a $\mathbf{Z}$ -module. More generally, by Galois theory, $E(\bar{F})$ is a $\mathbf{Z}\mathrm{Gal}(\bar{F}/F)$ -module. In particular, by Section 2.1, we can consider the cohomology groups $H^*(\mathrm{Gal}(\bar{F}/F), E(\bar{F}))$, denoted by $H^1(F, E)$. More generally, for any $\mathrm{Gal}(\bar{F}/F)$ -module M , we denote by $H^*(F, M)$ the cohomology groups $H^*(\mathrm{Gal}(\bar{F}/F), M)$.

Definition 3.1.2 (Kummer morphism). Let E be an elliptic curve defined over F and M/F be an algebraic extension. The Kummer morphism

$$\kappa: E(M) \otimes \mathbf{Q}/\mathbf{Z} \rightarrow H^1(M, E(\bar{F})_{\mathrm{tors}})$$

is defined such that for any $\alpha = P \otimes (m/n + \mathbf{Z})$ from $E(M) \otimes \mathbf{Q}/\mathbf{Z}$, $\kappa(\alpha)$ is the cohomology class of the cocycle $\sigma \mapsto Q^\sigma - Q$ for some $Q \in E(\bar{F})$ such that $nQ = mP$.

Remark. Let E , M/F , and α be defined as in Definition 3.1.2. Let Q and R be two points from $E(\bar{F})$ such that $mQ = nP = mR$.

First, the map $\sigma \mapsto Q^\sigma - Q$ is indeed a map from $\mathrm{Gal}(\bar{F}/M)$ to $E(\bar{F})_{\mathrm{tors}}$ because, for any $\sigma \in \mathrm{Gal}(\bar{F}/M)$, $m(Q^\sigma - Q) = (nP)^\sigma - nP = 0$ since nP is fixed by $\mathrm{Gal}(\bar{F}/M)$.

Then, we have for any $\sigma \in \text{Gal}(\bar{F}/M)$

$$(Q^\sigma - Q) - (R^\sigma - R) = (Q - R)^\sigma - (Q - R)$$

which can be written $d(Q - R)(\sigma)$ because $m(Q - R) = 0$ so $Q - R$ is torsion.

Hence, κ is well-defined.

We recall that E is uniquely determined (up to a change of coordinates) by a cubic equation called minimal Weierstrass equation (see Silverman [15, Chapter VII]).

Let p be a prime number. We denote by $f_{\mathfrak{p}}$ the residue field of $F_{\mathfrak{p}}$.

Definition 3.1.3. Let $\mathfrak{p}$ be a prime ideal over p . The reduction $\tilde{E} = E \bmod \mathfrak{p}$ of a minimal Weierstrass equation defining E is

- *good* if $\tilde{E}$ is smooth, i.e., $\tilde{E}$ is an elliptic curve over $\mathcal{O}_F/\mathfrak{p}$,
- *ordinary* if $\tilde{E}(\bar{f}_{\mathfrak{p}})[p] \neq 0$,
- *multiplicative* if $\tilde{E}$ has a nodal singularity.

Remark. In the above definition, to reduce a point P of $E(F)$ modulo $\mathfrak{p}$, one should write P as $[x : y : z]$ in projective coordinates where x, y and z are algebraic integers not all belonging to $\mathfrak{p}$. This is always possible, by multiplying the coordinates of P by a suitable element of F .

Let us recall that the formal group law associated to E is a formal group law over $\mathcal{O}_{F_{\mathfrak{p}}}$, i.e., a power series $\mathcal{F}(X, Y)$ with coefficients in $\mathcal{O}_{F_{\mathfrak{p}}}$ such that

- $\mathcal{F}(X, Y) = X + Y + \mathcal{G}(X, Y)$ where terms in $\mathcal{G}$ have order at least two,
- $\mathcal{F}(X, \mathcal{F}(Y, Z)) = \mathcal{F}(\mathcal{F}(X, Y), Z)$,
- $\mathcal{F}(X, Y) = \mathcal{F}(Y, X)$,
- There is a unique power series $\mathcal{I}(X)$ such that $\mathcal{F}(X, \mathcal{I}(X)) = 0$,
- $\mathcal{F}(X, 0) = X$.

We denote by $\mathcal{F}(\mathfrak{m})$ the maximal ideal $\mathfrak{m}$ of $\mathcal{O}_{F_{\mathfrak{p}}}$ endowed with the group law induced by $\mathcal{F}$. The p -adic logarithm establishes an isomorphism between $\mathcal{F}(\mathfrak{m})$ and $\mathfrak{m}$ (see Silverman [15, Chapter IV, Proposition 5.2]). Besides, we have the following exact sequence (see Silverman [15, Chapter VII, Proposition 2.2]):

$$0 \longrightarrow \mathcal{F}(\mathfrak{m}) \longrightarrow E(F_{\mathfrak{p}}) \longrightarrow E(f_{\mathfrak{p}}) \longrightarrow 0$$

Let m be an integer. Let us denote by $E[m]$ the group $E(\bar{F})[m]$ for short. Since $\bar{F}$ can be embedded in $\mathbf{C}$ and $E(\mathbf{C})$ is a torus, we have:

$$E[m] \cong \frac{\mathbf{Z}}{m\mathbf{Z}} \times \frac{\mathbf{Z}}{m\mathbf{Z}}.$$

Besides, $\text{Gal}(\overline{F}/F)$ acts on $E[m]$. Therefore, we have a representation

$$\text{Gal}(\overline{F}/F) \rightarrow \text{Aut}(E[m]) \cong \mathcal{GL}_2(\mathbf{Z}/m\mathbf{Z}).$$

We introduce the Tate-module that gathers all the p -power representations built as above when m is a p -power.

Definition 3.1.4. The p -adic Tate-module of E/F is

$$T_p(E) = \varprojlim_n E[p^n],$$

the inverse limit being taken with respect to the multiplication by p .

By construction, $T_p(E)$ is isomorphic to $\mathbf{Z}_p \times \mathbf{Z}_p$ and we have a continuous representation (with respect to the p -adic topology) $\text{Gal}(\overline{F}/F) \rightarrow \text{Aut}(T_p(E))$. Then, one can build an embedding $\text{Aut}(T_p(E)) \hookrightarrow \mathcal{GL}_2(\mathbf{Q}_p)$ by choosing a $\mathbf{Z}_p$ -basis for $T_p(E)$. Hence, we have a representation in a field of characteristic 0. In order to make an intrinsic definition, we consider the representation space $V_p(E) = T_p(E) \otimes \mathbf{Q}_p$.

The Weil pairing is a map

$$e: T_p(E) \times T_p(E) \longrightarrow \mathbf{Z}_p(1)$$

that is bilinear, alternating, non-degenerate and Galois invariant, i.e., for any $\sigma \in \text{Gal}(\overline{F}/F)$ and any $(x, y) \in T_p(E) \times T_p(E)$, we have $e(x, y)^\sigma = e(x^\sigma, y^\sigma)$ (see Silverman [15, Chapter III, Proposition 8.3]). In particular, e induces an isomorphism between $T_p(E) \wedge T_p(E)$ and $\mathbf{Z}_p(1)$ (see Lang [16, Chapter XIX] for an introduction to the alternating product $\wedge$).

3.2 Kummer theory for elliptic curves

In this section, we introduce Kummer theory for elliptic curves. See in particular Silverman [15, Chapter VIII] for a complete introduction to the subject. Let M/F be a Galois extension. We suppose that M/F is either finite or a $\mathbf{Z}_p$ -extension. In addition, $\mathcal{M}_L$ denotes the set of places of L for any field L and p denotes a prime number. Moreover, we fix an integer m .

Let us consider the exact sequence

$$0 \longrightarrow E(\overline{F})[m] \longrightarrow E(\overline{F}) \xrightarrow{[m]} E(\overline{F}) \longrightarrow 0$$

The group $\text{Gal}(\overline{F}/M)$ acts on $E(\overline{F})$ since M/F is algebraic. By Galois cohomology long exact sequence, we can build the following exact sequence, noticing that $E(M) = H^0(M, E(\overline{F}))$ since the Galois group fixes exactly M -points of the elliptic curve.

$$\begin{array}{ccccccc} 0 & \longrightarrow & E(M)[m] & \longrightarrow & E(M) & \xrightarrow{[m]} & E(M) \\ & & & & & & \searrow \\ & & & & & & \delta \\ & & & & & & \searrow \\ & & & & & & \longrightarrow H^1(M, E(\overline{F})[m]) \longrightarrow H^1(M, E(\overline{F})) \xrightarrow{[m]} H^1(M, E(\overline{F})) \end{array}$$

where δ is defined by $\delta(P) : \sigma \mapsto Q^\sigma - Q$ where Q is any element of $E(\overline{F})$ such that $P = mQ$. Comparing to Definition 3.1.2, we get $\delta = \kappa(\cdot, m + \mathbf{Z})$. Then, from the above exact sequence, we can extract the following sequence by noticing — by exactness — that the kernel of δ is the image of $[m]$ (seen as an endomorphism of $E(M)$) and that the image of the penultimate morphism is the kernel of $[m]$ (seen as an endomorphism of $H^1(M, E(\overline{F}))$):

$$0 \longrightarrow \frac{E(M)}{mE(M)} \xrightarrow{\kappa} H^1(M, E[m]) \longrightarrow H^1(M, E)[m] \longrightarrow 0 \quad (3.1)$$

The map κ above is actually the one induced by the Kummer morphism from Definition 3.1.2 by identifying $E(M)/mE(M)$ with $E(M) \otimes \mathbf{Z}/m\mathbf{Z}$ and noticing that $\mathbf{Q}/\mathbf{Z}$ is the direct limit of $\mathbf{Z}/n\mathbf{Z}$ for $n \in \mathbf{N}$.

The diagram (3.1) is called Kummer sequence. Notice that we write E instead of $E(\overline{F})$ for short in the diagram (3.1). We will omit the field over which points are taken whenever it is clear from the context.

Let us consider ν , a valuation on M . We denote by G_ν the decomposition group associated to ν . In other words, G_ν is the subgroup of $\text{Gal}(\overline{F}/M)$ that preserves ν . Moreover, G_ν is canonically isomorphic to $\text{Gal}(\overline{F}/M_\nu)$ (see Neukirch [17, chapter II, proposition 9.6] for a proof of this fact). Building the Kummer sequence for any completion M_η of M , we can build the following commutative diagram where horizontal arrows are exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \frac{E(M)}{mE(M)} & \xrightarrow{\kappa} & H^1(M, E[m]) & \longrightarrow & H^1(M, E)[m] \longrightarrow 0 \\ & & \downarrow & & \downarrow & \searrow \theta & \downarrow \\ 0 & \rightarrow & \prod_{\eta \in \mathcal{M}_M} \frac{E(M_\eta)}{mE(M_\eta)} & \rightarrow & \prod_{\eta \in \mathcal{M}_M} H^1(M_\eta, E[m]) & \rightarrow & \prod_{\eta \in \mathcal{M}_M} H^1(M_\eta, E)[m] \rightarrow 0 \end{array}$$

where the first vertical arrow is given by the inclusions $E(M) \subset E(M_\eta)$ and the others are restriction morphisms given by the inclusions $G_\eta \subset \text{Gal}(\overline{F}/M)$.

Let us show that the above diagram is indeed commutative. The commutativity of the right square is straightforward. Now, let $P + mE(M)$ be an element of $E(M)/mE(M)$. Then, its image by the descending arrow is $(P + mE(M_\eta))_{\eta \in \mathcal{M}_M}$ which is itself sent to $(\kappa_\eta(P))_{\eta \in \mathcal{M}_M}$, where for any $\eta \in \mathcal{M}_M$ and any $\sigma \in G_\eta$, $\kappa_\eta(P)(\sigma) = Q_\eta^\sigma - Q_\eta$ and Q_η is an element of $E(\overline{F}_\eta)$ such that $mQ_\eta = P$. On the other hand, let Q be a point of $E(\overline{F})$ such that $mQ = P$. Then, we have for any $\sigma \in G_\nu$

$$(Q^\sigma - Q) - (Q_\nu^\sigma - Q_\nu) = (Q - Q_\nu)^\sigma - (Q - Q_\nu).$$

Since $mQ = P = mQ_\nu$, we have $m(Q - Q_\nu) = 0$ and the above element can be rewritten $d(Q - Q_\nu)(\sigma)$. Hence, the two images of $P + mE(M)$ differ (component-wise) by a coboundary and are thereby equal.

Definition 3.2.1 (m -th Selmer group). We define the m -th Selmer group of E over M as

$$\mathrm{Sel}_E^m(M) = \ker \left\{ H^1(M, E[m]) \rightarrow \prod_{\eta \in \mathcal{M}_M} H^1(M_\eta, E) \right\}.$$

Equivalently, $\mathrm{Sel}_E^m(M)$ is the kernel of θ .

Definition 3.2.2 (Selmer group). We define the Selmer group of E over M as

$$\mathrm{Sel}_E(M) = \varinjlim_n \mathrm{Sel}_E^n(M).$$

By exactness and commutativity, it is clear that the image of κ is included in $\mathrm{Sel}_E^m(M)$. Suppose we have an element φ in $\mathrm{Sel}_E^m(M) \setminus \mathrm{Im}(\kappa)$. In particular, by exactness, φ is not in the kernel of the horizontal arrow coming from $H^1(M, E[m])$. However, $(\varphi_\eta)_{\eta \in \mathcal{M}_M}$, its image given by restrictions to decomposition groups, is in the kernel of the lower right arrow. By exactness, each φ_η is given by $\kappa_\eta(P_\eta)$. Thus, each localization of φ comes from a point, but φ itself cannot be written as $\kappa(P)$ for any point $P \in E(F)$. This can be interpreted as a failure of the local-global principle in the light of the isomorphism between $H^1(M, E)$ and the Weil-Châtelet group of M -equivalence classes of homogeneous spaces, i.e., the group of smooth curves C defined over M such that E acts simply and transitively on C , quotiented by M -defined E -module isomorphism (see Silverman [15, Chapter X, Theorem 3.6] for details). In other words, interpreting the image of φ in $H^1(M, E)[m]$ as a homogeneous space C , we conclude that $C(M_\eta)$ is non-empty for any valuation η but $C(M)$ is empty. For instance, Selmer [18] showed that the equation

$$3x^3 + 4y^3 + 5z^3 = 0$$

has a solution in $\mathbf{P}^2(\mathbf{R})$ and $\mathbf{P}^2(\mathbf{Q}_p)$ for any prime p but not in $\mathbf{P}^2(\mathbf{Q})$.

In order to measure how much the local-global principle fails, we introduce the Shafarevich-Tate group.

Definition 3.2.3 (Shafarevich-Tate group). We define the Shafarevich-Tate group of E over M as

$$\mathrm{III}_E(M) = \ker \left\{ H^1(M, E) \rightarrow \prod_{\eta \in \mathcal{M}_M} H^1(M_\eta, E) \right\}.$$

The latter measures how much $\mathrm{Sel}_E^m(M)$ differs from $\mathrm{Im}(\kappa)$ as illustrated by the following exact sequence:

$$0 \longrightarrow \frac{E(M)}{mE(M)} \xrightarrow{\kappa} \mathrm{Sel}_E^m(M) \longrightarrow \mathrm{III}_E(M)[m] \longrightarrow 0 \quad (3.2)$$

In light of the discussion above, the bigger $\mathrm{III}_E(M)$ is, the wider is the failure of the local-global principle. The following conjecture — usually attributed to Tate and Shafarevich even though it might originate from Lang as mentioned by Cassels [19] — suggests that this failure might be not too dramatic:

Conjecture 3.2.4. *The Shafarevich-Tate group $\text{III}_E(F)$ is finite.*

Taking the direct limit in (3.1) (by identifying $E(M)/nE(M)$ with the group $E(M) \otimes \mathbf{Z}/n\mathbf{Z}$ for any integer n), we obtain the following exact sequence

$$0 \rightarrow E(M) \otimes (\mathbf{Q}/\mathbf{Z}) \xrightarrow{\kappa} H^1(M, E_{\text{tors}}) \rightarrow H^1(M, E)_{\text{tors}} \rightarrow 0 \quad (3.3)$$

Notice that in the diagram (3.3), κ is exactly the Kummer morphism defined in Definition 3.1.2.

The diagram (3.3) follows from taking the direct limit of (3.1). Indeed, let m and n be two integers. Suppose $m|n$. We need to check the commutativity of

$$\begin{array}{ccc} E(M) \otimes \frac{\mathbf{Z}}{m\mathbf{Z}} & \longrightarrow & H^1(M, E[m]) \\ \downarrow & & \downarrow \\ E(M) \otimes \frac{\mathbf{Z}}{n\mathbf{Z}} & \longrightarrow & H^1(M, E[n]) \end{array}$$

where the right arrows are given by multiplication by n/m . Let $P \otimes (l + m\mathbf{Z})$ be an element of $E(M) \otimes \mathbf{Z}/m\mathbf{Z}$. It is sent to $P \otimes (n/m)l + n\mathbf{Z}$ by the left vertical arrow. Then, it is sent to the map cohomology class of $\sigma \mapsto Q^\sigma - Q$ where Q is a point of $E(\overline{F})$ such that $nQ = (n/m)lP$. On the other hand, $P \otimes (l + m\mathbf{Z})$ is sent by the horizontal arrow to the cohomology class of $\sigma \mapsto R^\sigma - R$ where R is a point of $E(\overline{F})$ such that $mR = lP$. This last cohomology class is sent to $\sigma \mapsto (n/m)R^\sigma - (n/m)R$. Now, for any $\sigma \in \text{Gal}(\overline{F}/M)$, we have

$$((n/m)R^\sigma - (n/m)R) - (Q^\sigma - Q) = ((n/m)R - Q)^\sigma - ((n/m)R - Q)$$

which can be rewritten $d((n/m)R - Q)$ since $n((n/m)R - Q) = 0$.

Thus, since $\mathbf{Q}/\mathbf{Z} = \varinjlim_n \mathbf{Z}/n\mathbf{Z}$ and $E_{\text{tors}} = \varinjlim_n E[n]$, the result follows from Proposition 2.1.3.

Similarly, we can build the Kummer morphism (still denoted κ)

$$\kappa: E(M) \otimes (\mathbf{Q}_p/\mathbf{Z}_p) \rightarrow H^1(M, E[p^\infty])$$

thanks to a direct limit of (3.1), this time by noting that $\mathbf{Q}_p/\mathbf{Z}_p = \varinjlim_n \mathbf{Z}/p^n\mathbf{Z}$ and $E[p^\infty] = \varinjlim_n E[p^n]$. Besides, we denote by κ_η the Kummer morphism from $E(M_\eta) \otimes (\mathbf{Q}_p/\mathbf{Z}_p)$ to $H^1(M_\eta, E[p^\infty])$ built as above with M_η instead of M , for any valuation η on M .

Doing the same work as before, we can now introduce the following definitions:

Definition 3.2.5. The p -primary subgroup of $\text{Sel}_E(M)$ is

$$\text{Sel}_E(M)_p = \ker \left\{ H^1(M, E[p^\infty]) \rightarrow \prod_{\eta \in \mathcal{M}_M} H^1(M_\eta, E[p^\infty]) / \text{Im}(\kappa_\eta) \right\}.$$

Definition 3.2.6. The p -primary subgroup of $\text{III}_E(M)$ is

$$\text{III}_E(M)_p = \text{Sel}_E(M)_p / \text{Im}(\kappa).$$

From now on, suppose M/F finite.

A theorem of Faltings [20, 21] states that E is determined up to an isogeny of degree prime to p by the $\text{Gal}(\bar{F}/F)$ -module $V_p(E)/T_p(E)$. In addition, we have

$$\frac{V_p(E)}{T_p(E)} \cong \frac{T_p(E) \otimes \mathbf{Q}_p}{T_p(E) \otimes \mathbf{Z}_p} \cong T_p(E) \otimes \frac{\mathbf{Q}_p}{\mathbf{Z}_p} \cong \varinjlim_n \frac{T_p(E)}{p^n T_p(E)} \cong \varinjlim_n E[p^n] \cong E[p^\infty].$$

Besides, $\text{Sel}_E(M)_p$ is invariant by such isogenies. Hence, we naturally aim to define $\text{Sel}_E(M)_p$ only using $E[p^\infty]$. According to Definition 3.2.5, we need to describe the image of κ_η for $\eta \in \mathcal{M}_M$. This can be done under the assumption that E has good, ordinary or multiplicative reduction at all primes of F over p .

Proposition 3.2.7. *Then, for any valuation $\eta \in \mathcal{M}_M$, if $\eta \nmid p$ then $\text{Im}(\kappa_\eta) = 0$, otherwise $\text{Im}(\kappa_\eta) = (\mathbf{Q}_p/\mathbf{Z}_p)^{[F_p:\mathbf{Q}_p]}$.*

Proof. Let η be a valuation on M . First, suppose $\eta|q$ for some prime q . In this case, the kernel of the reduction map $\pi: E(M_\eta) \rightarrow E(m_\eta)$, where m_η is the residue field of M_η , is the formal group of E . Moreover, the q -adic logarithm identifies this formal group with the additive group given by the maximal ideal of $\mathcal{O}_{M_\eta}$. This last is isomorphic to $\mathbf{Z}_q^{[M_\eta:\mathbf{Q}_q]}$. Hence, $E(M_\eta)/\mathbf{Z}_q^{[M_\eta:\mathbf{Q}_q]}$ is finite and thereby, the $\mathbf{Z}_q$ -rank of $E(M_\eta)$ is $[M_\eta : \mathbf{Q}_q]$. In other words, $E(M_\eta)$ is isomorphic to $\mathbf{Z}_q^{[M_\eta:\mathbf{Q}_q]} \times U$ where U is a finite group.

Notice that

$$\mathbf{Z}_q \otimes \frac{\mathbf{Q}_p}{\mathbf{Z}_p} = \varinjlim_n \left(\mathbf{Z}_q \otimes \frac{\mathbf{Z}}{p^n \mathbf{Z}} \right) = \varinjlim_n \left(\frac{\mathbf{Z}_q}{p^n \mathbf{Z}_q} \right).$$

Hence, since p is invertible in $\mathbf{Z}_q$ when $p \neq q$, we have $\mathbf{Z}_q \otimes (\mathbf{Q}_p/\mathbf{Z}_p) = 0$. Otherwise, the above equation yields $\mathbf{Z}_p \otimes (\mathbf{Q}_p/\mathbf{Z}_p) \cong (\mathbf{Q}_p/\mathbf{Z}_p)$ since we have $\mathbf{Z}_p/p^n \mathbf{Z}_p \cong \mathbf{Z}/p^n \mathbf{Z}$ for any integer n . Moreover, since U is torsion and $\mathbf{Q}_p/\mathbf{Z}_p$ is divisible, we have $U \otimes (\mathbf{Q}_p/\mathbf{Z}_p) = 0$. Hence, $\text{Im}(\kappa_\eta)$ is zero whenever $p \neq q$ and $\text{Im}(\kappa_\eta) = (\mathbf{Q}_p/\mathbf{Z}_p)^{[M_\eta:\mathbf{Q}_q]}$ otherwise.

In the case $\eta|\infty$, we have $E(M_\eta) \cong (\mathbf{R}/\mathbf{Z})^{[M_\eta:\mathbf{R}]} \times U$ where U is finite (the complex case follows from the identification of $E(\mathbf{C})$ with a torus, for the real case, see Silverman [22, Chapter V, Corollary 2.3.1]) Since $\mathbf{R}/\mathbf{Z}$ is divisible, we get $E(M_\eta) \otimes (\mathbf{Q}_p/\mathbf{Z}_p) = 0$.

□

Let v be a prime of F lying above p . Suppose E has good, ordinary reduction at v . Let us denote by $\tilde{E}$ the reduction of E modulo v . We denote by C_v the kernel of the reduction map r from $E[p^\infty]$ to $\tilde{E}[p^\infty]$. As mentioned earlier, $E[p^n]$ is isomorphic to $(\mathbf{Z}/p^n \mathbf{Z})^2$ for any integer n so $E[p^\infty]$, as a direct limit, is isomorphic to $(\mathbf{Q}_p/\mathbf{Z}_p)^2$. Furthermore, since E has ordinary reduction at v , we have $\tilde{E}[p^\infty] \cong \mathbf{Q}_p/\mathbf{Z}_p$ (see Silverman [15, Chapter V, Theorem 3.1]). Since

the $\mathbf{Z}_p$ -corank of C_v is $\text{corank}_{\mathbf{Z}_p}(E[p^\infty]) - \text{corank}_{\mathbf{Z}_p}(\tilde{E}[p^\infty])$ which is 1. Hence, $C_v \cong \mathbf{Q}_p/\mathbf{Z}_p$.

Moreover, C_v is G_v -invariant (in other words, it is a G_v -module). Indeed, let $P \in C_v$ and $\sigma \in G_v$. Then, we have $r(P) = [a + \mathfrak{p} : b + \mathfrak{p} : c + \mathfrak{p}]$ where $[a : b : c]$ is the neutral point of E and $\mathfrak{p}$ is the ideal associated to v . In particular, we have $r(P^\sigma) = r(P)$ since G_v preserves $\mathfrak{p}$ and in particular preserves $\mathbf{Q}$ (remember that the neutral point is rational). Moreover, $E[p^\infty]/C_v$ is isomorphic to $\tilde{E}[p^\infty]$ and is thereby unramified. Hence, $E[p^\infty]/C_v$ is the maximal unramified quotient of $\tilde{E}[p^\infty]$.

By Galois cohomology long exact sequence, we get the following exact sequence:

$$\begin{array}{c} E(M_\eta)[p^\infty] \longrightarrow \tilde{E}(m_\eta)[p^\infty] \longrightarrow \\ \delta \downarrow \\ \longrightarrow H^1(M_\eta, C_v) \xrightarrow{\lambda_\eta} H^1(M_\eta, E[p^\infty]) \longrightarrow H^1(M_\eta, \tilde{E}[p^\infty]) \end{array} \quad (3.4)$$

where m_η denotes the residue field of M_η .

Proposition 3.2.8. *For any $\eta \in \mathcal{M}_M$, we have $\text{Im}(\kappa_\eta) = \text{Im}(\lambda_\eta)_{\text{div}}$, where $\text{Im}(\lambda_\eta)_{\text{div}}$ denotes the maximal divisible subgroup of $\text{Im}(\lambda_\eta)$.*

Proof. Let $\eta \in \mathcal{M}_M$.

First, note that $\text{Im}(\kappa_\eta)$ and $\text{Im}(\lambda_\eta)$ are p -primary groups and $\text{Im}(\kappa_\eta)$ is divisible since $\mathbf{Q}_p/\mathbf{Z}_p$ is divisible. In addition, by Proposition 3.2.7, $\text{Im}(\kappa_\eta)$ has $\mathbf{Z}_p$ -corank equal to $[M_\eta : \mathbf{Q}_p]$. Hence, it suffices to prove $\text{Im}(\kappa_\eta) \subset \text{Im}(\lambda_\eta)$ and that the $\mathbf{Z}_p$ -corank of $\text{Im}(\lambda_\eta)$ is $[M_\eta : \mathbf{Q}_p]$.

Indeed, if $\text{Im}(\kappa_\eta)$ is included in $\text{Im}(\lambda_\eta)$ then it is included in $\text{Im}(\lambda_\eta)_{\text{div}}$ since it is divisible. Then, suppose this first inclusion true and the $\mathbf{Z}_p$ -corank of $\text{Im}(\kappa_\eta)$ and $\text{Im}(\lambda_\eta)$ to be equal. By Proposition 3.2.7, this corank is $[M_\eta : \mathbf{Q}_p]$. Now, consider the inclusion map $\iota : \text{Im}(\kappa_\eta) \hookrightarrow \text{Im}(\lambda_\eta)_{\text{div}}$. Hence, ι induces an embedding of $(\mathbf{Q}_p/\mathbf{Z}_p)^{[M_\eta : \mathbf{Q}_p]}$ in itself according to Proposition 2.2.6. In addition, by Corollary 2.2.7, ι is an isomorphism. Hence, the inclusion is actually an equality.

First, let us consider $c \in \text{Im}(\kappa_\eta)$. According to diagram (3.4), we need to show that c is in the kernel of the map $H^1(M_\eta, E[p^\infty]) \rightarrow H^1(M_\eta, \tilde{E}[p^\infty])$. We denote by f_v the residue field of F_v . Let φ be a cocycle representing c . By definition of κ_η , there is $Q \in E(\overline{F_v})$ such that

$$\varphi : \begin{array}{ccc} G_\eta & \longrightarrow & E(\overline{F_v})[p^\infty] \\ \sigma & \longmapsto & Q^\sigma - Q \end{array}.$$

Let us denote by $\tilde{\varphi}$ the cocycle from G_η to $\tilde{E}(\overline{f_v})[p^\infty]$ induced by φ . In other words, denoting by $\tilde{Q}$ the point Q reduced to $\tilde{E}(\overline{f_v})$, $\tilde{\varphi}$ maps $\sigma \in G_\eta$ to $\tilde{Q}^\sigma - \tilde{Q}$. Let us denote by $\tilde{c}$ the cohomology class of $\tilde{\varphi}$ in $H^1(M_\eta, \tilde{E}[p^\infty])$. Let ω be the (finite) order of $\tilde{Q}$ and n_σ be the p -adic value of the order of $\tilde{\varphi}(\sigma)$ for any $\sigma \in G_\eta$.

We have $\omega\tilde{\varphi} = 0$ so for any $\sigma \in G_\eta$, $p^{n\sigma}$ divides ω . In particular, if n denotes the p -adic value of ω , then $p^n\tilde{\varphi} = 0$. This implies, by definition of $\tilde{\varphi}$, that p^nQ is fixed by G_η and thereby belongs to $\tilde{E}(f_v)$. Hence, by definition of $\tilde{E}$, since v lies above p , p^nQ is zero. Therefore, ω divides p^n . This shows that $\omega = p^n$. Thus, $\tilde{\varphi}$ is a coboundary and thereby, $\tilde{c} = 0$.

Now, let us compute the $\mathbf{Z}_p$ -corank of $\text{Im}(\lambda_\eta)$. According to the diagram (3.4), $\ker(\lambda_\eta)$ is finite as the image of a homomorphism whose domain is finite. Hence, the $\mathbf{Z}_p$ -corank of $\text{Im}(\lambda_\eta)$ and of $H^1(M_\eta, C_v)$ coincide according to Proposition 2.2.8. As discussed before, C_v is isomorphic to $\mathbf{Q}_p/\mathbf{Z}_p$ and G_v acts on C_v . Hence, $\mathbf{Q}_p/\mathbf{Z}_p$ inherits an action $G_v \rightarrow \mathbf{Z}_p^\times$, where $\mathbf{Z}_p^\times$ is the automorphism group of $\mathbf{Q}_p/\mathbf{Z}_p$ as the projective limit of $(\mathbf{Z}/p^n\mathbf{Z})^\times$. For any continuous homomorphism $\psi: G_v \rightarrow \mathbf{Z}_p^\times$, let us denote by $(\mathbf{Q}_p/\mathbf{Z}_p)(\psi)$ the group $\mathbf{Q}_p/\mathbf{Z}_p$ endowed with the G_v -module structure induced by ψ . Now, we need the following lemma to conclude:

Lemma 3.2.9. *Let $\psi: G_v \rightarrow \mathbf{Z}_p^\times$ be a continuous morphism. The $\mathbf{Z}_p$ -corank of $H^1(M_\eta, (\mathbf{Q}_p/\mathbf{Z}_p)(\psi))$ is $[M_\eta : \mathbf{Q}_p] + \delta$ where $\delta = 1$ if $\psi|_{G_\eta}$ is trivial or the cyclotomic character and $\delta = 0$ otherwise.*

Proof. First, suppose $\psi|_{G_\eta}$ trivial. Then, any crossed morphism is just a morphism. Moreover, since $\mathbf{Q}_p/\mathbf{Z}_p$ is abelian, any morphism from G_η to $\mathbf{Q}_p/\mathbf{Z}_p$ factors through the maximal abelian subgroup of G_η which is $\text{Gal}(M_\eta^{\text{ab}}/M_\eta)$ where M_η^{ab} denotes the maximal abelian extension of M_η . Hence, we have

$$H^1(M_\eta, (\mathbf{Q}_p/\mathbf{Z}_p)(\psi)) = \text{Hom}(\text{Gal}(M_\eta^{\text{ab}}/M_\eta), \mathbf{Q}_p/\mathbf{Z}_p).$$

Moreover, $\text{Gal}(M_\eta^{\text{ab}}/M_\eta)$ (see Hazewinkel [23, Theorem 7] for instance) is isomorphic to

$$\mathbf{Z}_p^{[M_\eta:\mathbf{Q}_p]} \times \hat{\mathbf{Z}} \times (M_\eta^\times)_{\text{tors}}$$

where $\hat{\mathbf{Z}}$ denotes the profinite completion of $\mathbf{Z}$, i.e., $\hat{\mathbf{Z}}$ is the projective limit of the $\mathbf{Z}/n\mathbf{Z}$'s for $n \in \mathbf{N}$. In particular, $\hat{\mathbf{Z}}$ is isomorphic to $\prod_q \mathbf{Z}_q$ (where the product is taken over prime numbers). Indeed, this is a consequence of the Chinese remainder theorem. Since $(M_\eta^\times)_{\text{tors}}$ is finite and there is no embedding of $\mathbf{Z}_p$ in $\mathbf{Z}_q$ for any prime q distinct from p , this gives the result in this case.

Now, suppose $\psi|_{G_\eta}$ is the cyclotomic character. By definition, $(\mathbf{Q}_p/\mathbf{Z}_p)(\psi)$ is isomorphic to μ_{p^∞} , the set of p -power roots of unity inside F . The exact sequence (where $[p^m]$ denotes the raising to the p^m -th power)

$$0 \longrightarrow \overline{F}^\times[p^m] \longrightarrow \overline{F}^\times \xrightarrow{[p^m]} \overline{F}^\times \longrightarrow 0$$

yields through Galois cohomology long sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & M_\eta^\times[p^m] & \longrightarrow & M_\eta^\times & \xrightarrow{[p^m]} & M_\eta^\times \\ & & & & & & \downarrow \delta \\ & & & & & & H^1(M_\eta, \overline{F}^\times[p^m]) \longrightarrow H^1(M_\eta, \overline{F}^\times) \xrightarrow{[p^m]} H^1(M_\eta, \overline{F}^\times) \end{array}$$

By Theorem 2.1.6 $H^1(M_\eta, \overline{F}^\times)$ is trivial. Hence, δ induces an isomorphism κ — which is actually a Kummer morphism — from $M_\eta^\times \otimes \mathbf{Z}/p^m \mathbf{Z}$ to $H^1(M_\eta, \overline{F}^\times[p^m])$. Taking the direct limit, we deduce that κ induces an isomorphism between $M_\eta^\times \otimes \mathbf{Q}_p/\mathbf{Z}_p$ and $H^1(M_\eta, \mu_{p^\infty})$.

Hence, we are reduced to calculating the $\mathbf{Z}_p$ -corank of $M_\eta^\times \otimes \mathbf{Q}_p/\mathbf{Z}_p$. First, let us recall that

$$M_\eta^\times \cong \mathbf{Z} \times (\mathbf{Z}_p^\times)^{[M_\eta : \mathbf{Q}_p]} \cong \mathbf{Z} \times \left(\left(\frac{\mathbf{Z}}{p\mathbf{Z}} \right)^\times \times \mathbf{Z}_p \right)^{[M_\eta : \mathbf{Q}_p]}.$$

Now, notice that

$$\mathbf{Z} \otimes \frac{\mathbf{Q}_p}{\mathbf{Z}_p} = \frac{\mathbf{Q}_p}{\mathbf{Z}_p}$$

and

$$\mathbf{Z}_p \otimes \frac{\mathbf{Q}_p}{\mathbf{Z}_p} = \varinjlim_n \left(\mathbf{Z}_p \otimes \frac{\mathbf{Z}}{p^n \mathbf{Z}} \right) = \varinjlim_n \frac{\mathbf{Z}_p}{p^n \mathbf{Z}_p} = \frac{\mathbf{Q}_p}{\mathbf{Z}_p}$$

Moreover, since $(\mathbf{Z}/p\mathbf{Z})^\times$ is finite and $\mathbf{Q}_p/\mathbf{Z}_p$ is divisible, their tensor product is trivial. Thus, the $\mathbf{Z}_p$ -corank of $M_\eta^\times \otimes \mathbf{Q}_p/\mathbf{Z}_p$ is $[M_\eta : \mathbf{Q}_p] + 1$ according to Proposition 2.2.4.

Now, let us exclude the cases where $\psi|_{G_\eta}$ is trivial or cyclotomic. Let M_∞/M_η be the field extension such that $\ker(\psi|_{G_\eta}) = \text{Gal}(\overline{M}_\eta/M_\infty)$. In particular, the image of $\psi|_{G_\eta}$, denoted by G , is isomorphic to $\text{Gal}(M_\infty/M_\eta)$. If ψ has finite order n then its image is finite since it is contained in the n -th roots of unity. Hence, if ψ has finite order, $M_\infty/\mathbf{Q}_p$ is finite and we can study the action of G on $\text{Gal}(M_\infty^{\text{ab}}/M_\infty)$. Suppose ψ has infinite order. If p is odd, then $\mathbf{Z}_p^\times$ is isomorphic to $(\mathbf{Z}/p\mathbf{Z})^\times \times \mathbf{Z}_p$ while $\mathbf{Z}_2^\times$ is isomorphic to $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}_2$. Hence, G is isomorphic to $\Delta \times H$ where Δ is finite of order dividing $p-1$ when p is odd and of order 1 or 2 otherwise, and H is isomorphic to $\mathbf{Z}_p$ — since ψ has infinite order. Let us write $C = (\mathbf{Q}_p/\mathbf{Z}_p)(\psi)$. By the inflation-restriction-sequence we have the following exact sequence:

$$0 \longrightarrow H^1(G, C) \longrightarrow H^1(M_\eta, C) \longrightarrow H^1(M_\infty, C)^G \longrightarrow H^2(G, C) \quad (3.5)$$

Let h be a topological generator of H , i.e., an element that generates a group dense in H . Seeing h as an endomorphism of C , we have $H^1(H, C) = C/(h-1)C$ according to Proposition 2.3.2.

Now, consider an element x in the kernel of $h-1$. If σ is in the preimage of h by ψ , then $(h-1) \cdot x = (\psi(\sigma)-1)x$. So, denoting by p^k the order of x , we deduce that $\psi(\sigma)-1$ belongs to $p^k \mathbf{Z}_p$. In particular, $h-1$ cancels elements of order less than or equal to p^k . Hence, its kernel is isomorphic to $\mathbf{Z}/p^k \mathbf{Z}$. Moreover, $\text{Im}(h-1)$ is divisible and p -primary as the image of a divisible p -primary group. Besides, the finiteness of $\ker(h-1)$ implies that $\text{Im}(h-1)$ is not trivial. So, by Corollary 2.2.7, $h-1$ is surjective and thereby, $C/(h-1)C = 0$. Hence, $H^1(H, C) = 0$. Combining this with the inflation-restriction sequence

$$0 \longrightarrow H^1(\Delta, C^H) \longrightarrow H^1(G, C) \longrightarrow H^1(H, C)^\Delta$$

gives an isomorphism between $H^1(G, C)$ and $H^1(\Delta, C^H)$. Now, if p is odd, then Δ has order prime with p and thereby the multiplication by $|\Delta|$ is an automorphism of $\mathbf{Q}_p/\mathbf{Z}_p$. Let φ be a cocycle from Δ to C^H . Then, when p is odd we can write

$$\alpha = \frac{1}{|\Delta|} \sum_{\delta \in \Delta} \varphi(\delta).$$

and the cocycle condition shows that $\varphi = d\alpha$ is a coboundary. Therefore, $H^1(\Delta, C^H)$ is trivial and so is $H^1(G, C)$ when p is odd. Suppose $p = 2$ and let τ be a generator of Δ . Then, $H^1(\Delta, C^H)$ is isomorphic to $\ker(\tau + 1)/\text{Im}(\tau - 1)$ according to Proposition 2.1.2. Since τ acts by multiplication by an element of $\mathbf{Z}_2^\times$ of order dividing 2, it is either the identity or the multiplication by -1 . In the first case, $H^1(\Delta, C^H)$ is isomorphic to $\mathbf{Z}/2\mathbf{Z}$ and in the second case it is trivial.

Besides, it is known that $H^2(H, C) = 0$ since H is a free pro- p group. As previously, when p is odd, this implies $H^2(G, C) = 0$ while it is a group of order at most 2 when $p = 2$.

Thus, from (3.5), we deduce that $H^1(M_\eta, C)$ and $H^1(M_\infty, C)^G$ are isomorphic when p is odd, and are pseudo-isomorphic through a map with kernel and cokernel of order at most 2 otherwise. In any case, they have the same $\mathbf{Z}_p$ -corank according to Proposition 2.2.8. By definition, any element of $H^1(M_\infty, C)^G$ is a G -module morphism from $\text{Gal}(\bar{M}_\eta/M_\infty)$ — on which G acts by conjugation — to C . Moreover, by commutativity of C , such a morphism factors through $\text{Gal}(M_\infty^{\text{ab}}/M_\infty)$, the maximal abelian subgroup of $\text{Gal}(\bar{M}/M_\infty)$. Hence, we are reduced to studying $\text{Hom}_G(\text{Gal}(M_\infty^{\text{ab}}/M_\infty), C)$. Note that $\text{Gal}(M_\infty^{\text{ab}}/M_\infty)$ is the projective limit of the $\text{Gal}(M'/M_\infty)$'s where M' varies through finite extensions of M_∞ contained in M_∞^{ab} . Then, since C is p -primary, for any such M' , a morphism from $\text{Gal}(M'/M_\infty)$ to C factors as a morphism from $\text{Gal}(M'/M_\infty)[p^\infty]$ to C . Hence, a morphism from $\text{Gal}(M_\infty^{\text{ab}}/M_\infty)$ to C factors through the maximal pro- p subgroup of $\text{Gal}(M_\infty^{\text{ab}}/M_\infty)$. Hence, we are reduced to studying $\text{Hom}_G(\text{Gal}(L_\infty/M_\infty), C)$ where L_∞ is the maximal abelian pro- p extension of M_∞ .

Let X denote the Galois group $\text{Gal}(L_\infty/M_\infty)$. Let Λ denote the Iwasawa algebra, i.e., $\Lambda = \mathbf{Z}_p[[H]]$. We are going to study X as a $\mathbf{Z}_p[[G]]$ -module. Notice that $\mathbf{Z}_p[[G]] = \Lambda[\Delta]$:

$$\mathbf{Z}_p[[G]] = \varprojlim_n \mathbf{Z}_p[(H/H_n) \times \Delta] = \varprojlim_n \mathbf{Z}_p[H/H_n][\Delta] = \mathbf{Z}_p[[H]][\Delta].$$

Let $H_n = H^{p^n}$ and $M_n = M_\infty^{H_n}$ for any integer n .

By Galois theory, we have that H_m is isomorphic to $\text{Gal}(L_\infty/M_m)/X$ and H_m acts on X by conjugation. Since, H_m is a free pro- p group, $H^2(H_m, X) = 0$. Hence, the following exact sequence splits:

$$1 \longrightarrow X \longrightarrow \text{Gal}(L_\infty/M_m) \longrightarrow H_m \longrightarrow 1$$

Consequently, we have $\text{Gal}(L_\infty/M_m) \cong X \rtimes H_m$. Therefore, the commutator of

$\text{Gal}(L_\infty/M_m)$ is

$$[X, X][X, H_m] \rtimes [H_m, H_m] \cong [X, H_m] \cong [H_m, X]$$

noticing that H_m and X are commutative. Now, for any $x \in X$, we have

$$[h^{p^m}, x] = h^{p^m} x h^{-p^m} x^{-1} = (h^{p^m} \cdot x)(-1 \cdot x) = (h^{p^m} - 1) \cdot x.$$

Since H_m is generated by h^{p^m} , we deduce that the commutator of $\text{Gal}(L_\infty/M_m)$ is isomorphic to $(h^{p^m} - 1)X$.

Let L_m be the maximal abelian extension of M_m contained in L_∞ . By Galois theory and the previous discussion, we have:

$$\text{Gal}(L_m/M_m) \cong \frac{X \rtimes H_m}{(h^{p^m} - 1)X \rtimes 1} \cong H_m \times \frac{X}{(h^{p^m} - 1)X}$$

Note that this holds because the action of H_m on $X/(h^{p^m} - 1)X$ is trivial by construction. Moreover, this Galois group is isomorphic to $\mathbf{Z}_p^{[M_m:\mathbf{Q}_p]+1} \times W_m$ where $W_m = M_m \cap \mu_{p^\infty}$. Indeed, it is the pro- p completion of $\text{Gal}(M_m^{\text{ab}}/M_m)$, i.e., of $\mathbf{Z}_p^{[M_m:\mathbf{Q}_p]} \times \widehat{\mathbf{Z}} \times (M_m^\times)_{\text{tors}}$ as mentioned earlier. Let $t = [M_0 : \mathbf{Q}_p]$. Since $\Delta = G/H$, we have by Galois theory $|\Delta| = [M_0 : M_\eta]$. Hence, we have $t = |\Delta|[M_\eta : \mathbf{Q}_p]$. Moreover, we have $[M_m : \mathbf{Q}_p] = p^m t$. Since H_m is isomorphic to $\mathbf{Z}_p$, this yields

$$\frac{X}{(h^{p^m} - 1)X} \cong \mathbf{Z}_p^{tp^m} \times W_m.$$

Now, let r denote the Λ -rank of X .

We have

$$\frac{\Lambda}{(h^{p^m} - 1)\Lambda} \cong \frac{\mathbf{Z}_p[[T]]}{((T+1)^{p^m} - 1)\mathbf{Z}_p[[T]]} \cong \mathbf{Z}_p^{p^n}.$$

Let us denote by Y the quotient $X/X_{\Lambda\text{-tors}}$. Then, Y is a submodule of finite index of Λ^r according to Theorem 2.3.7. In particular, we have the following exact sequence:

$$\frac{Y}{(h^{p^m} - 1)Y} \xrightarrow{f} \mathbf{Z}_p^{rp^m} \longrightarrow \text{coker}(f) \longrightarrow 0$$

where $\text{coker}(f)$ is a finite group. Let y be a non trivial element of the kernel of f . Then, y can be written as $(h^{p^m} - 1)\lambda$ where λ belongs to $\Lambda^r \setminus Y$. Note that, since Y has finite index in Λ^r , we can choose a finite set $\{\lambda_1, \dots, \lambda_s\}$ of representatives of Λ^r/Y . So, $\lambda = y'\lambda_i$ for some $y' \in Y$ and $i \in \{1, \dots, s\}$. In particular, since $(h^{p^m} - 1)y'$ is trivial in the quotient, this shows that $\ker(f)$ is finite. Thus, f is a pseudo-isomorphism and we deduce that $Y/(h^{p^m} - 1)Y$ and $\mathbf{Z}_p^{rp^m}$ have the same $\mathbf{Z}_p$ -corank. We deduce that $r = t$.

Suppose that M_∞ does not contain μ_{p^∞} . Hence, W_n has bounded order as n varies through $\mathbf{N}$. In particular, $(W_n)_{n \in \mathbf{N}}$ is eventually constant, equal to W_N for some $N \in \mathbf{N}$ great enough. Now, let us remark that $X_{\Lambda\text{-tors}}$ is the projective limit of the W_n 's taken over the norm maps (see Neukirch [17, Chapter IV,

Proposition 5.8]). Thus, any root $\zeta \in W_{2N} = W_N$ is sent by the norm map $W_{2N} \rightarrow W_N$ to $\zeta^{p^N} = 1$. In other words, the projective limit $X_{\Lambda\text{-tors}}$ is trivial. We deduce from this that X is included in Λ^t and has finite index in Λ^t . Hence, for some integer l large enough, $h^{p^l} - 1$ annihilates Λ^t/X . Thereby, we have

$$(h^{p^l} - 1)X \subset (h^{p^l} - 1)\Lambda^t \subset X \subset \Lambda^t$$

from which we deduce the following exact sequence:

$$0 \longrightarrow \frac{(h^{p^l}-1)\Lambda^t}{(h^{p^l}-1)X} \longrightarrow \frac{X}{(h^{p^l}-1)X} \xrightarrow{f} \frac{X}{(h^{p^l}-1)\Lambda^t} \longrightarrow 0$$

Let $x + (h^{p^l} - 1)X$ be a non-trivial element of the kernel of f . Then, $x = (h^{p^l} - 1)\lambda$ for some $\lambda \in \Lambda^t \setminus X$. Since X has finite index in Λ^t , the element $\lambda + X$ of Λ^t/X has finite order. In other words, $x + (h^{p^l} - 1)X$ has finite order. In addition, $X/(h^{p^l} - 1)\Lambda^t$ is a subgroup of $\Lambda^t/(h^{p^l} - 1)\Lambda^t$, itself isomorphic to $\mathbf{Z}_p^{p^l}$, a torsion-free group. Hence, the kernel of f is exactly the torsion subgroup of $X/(h^{p^l} - 1)X$. Now, since Λ^t is an integral domain, the multiplication by $h^{p^l} - 1$ is an injective map, from which it follows that $(h^{p^l} - 1)\Lambda^t/(h^{p^l} - 1)X$ is isomorphic to Λ^t/X . Thus, Λ^t/X is isomorphic to $(X/(h^{p^l} - 1)X)_{\text{tors}}$, i.e., to W where $W = M_{\infty}^{\times} \cap \mu_{p^{\infty}}$.

Otherwise, if M_{∞} contains $\mu_{p^{\infty}}$ then $X_{\Lambda\text{-tors}}$ is isomorphic to $\mathbf{Z}_p(1)$. Moreover, applying the above reasoning with $Y = X/X_{\Lambda\text{-tors}}$, we deduce that Λ^t/Y is isomorphic to $(Y/(h^{p^l} - 1)Y)_{\text{tors}}$, which is trivial. In other words, Y is free, i.e., isomorphic to Λ^t . Hence, the following exact sequence splits:

$$0 \longrightarrow \mathbf{Z}_p(1) \longrightarrow X \longrightarrow \Lambda^t \longrightarrow 0$$

In other words, X is isomorphic to $\Lambda^t \times \mathbf{Z}_p(1)$.

Besides, the action of Δ on X induces an action on Λ^t since Λ^t is the reflexive hull of $X/X_{\Lambda\text{-tors}}$ — in other words, Λ^t is isomorphic to $\text{Hom}(\text{Hom}(X, \Lambda), \Lambda)$ — and Δ preserves the Λ -torsion (see Bourbaki [24] for an introduction to reflexive hulls and Hartshorne [25, Theorem 1.12] for the above results). Suppose p odd. Then, recalling the isomorphism between $\mu_{p^{\infty}}$ and $\mathbf{Q}_p/\mathbf{Z}_p$, a character of Δ — that is, a morphism from Δ to $\mathbf{C}^{\times}$ — takes values in $\mathbf{Z}_p^{\times}$ since $|\Delta|$ is prime with p and the torsion subgroup of $\mathbf{Z}_p^{\times}$ is isomorphic to $(\mathbf{Z}/p\mathbf{Z})^{\times}$. Now, consider for any such character $\varphi \in \Delta^*$ the following element from $\mathbf{Z}_p[\Delta]$:

$$e_{\varphi} = \frac{1}{|\Delta|} \sum_{\delta \in \Delta} \varphi(\delta) \delta^{-1}.$$

Note that it is well-defined because the multiplication by $|\Delta|$ is an isomorphism of $\mathbf{Z}_p$. We have for any characters φ and ψ of Δ :

$$e_{\varphi} e_{\psi} = \begin{cases} e_{\varphi} & \text{if } \varphi = \psi, \\ 0 & \text{else.} \end{cases}$$

Moreover, we have

$$\sum_{\varphi \in \Delta^*} e_\varphi = 1.$$

Hence, we can write

$$\Lambda^t = \bigoplus_{\varphi \in \Delta^*} e_\varphi \Lambda^t.$$

Now, $e_\varphi \Lambda^t$ is isomorphic to $\Lambda^{[M_\eta, \mathbf{Q}_p]}$ for any character χ according to Gillard [26, Proposition 1]. This yields the $\Lambda[\Delta]$ -module isomorphism $\Lambda^t \cong \Lambda[\Delta]^{[M_\eta, \mathbf{Q}_p]}$. Suppose now $p = 2$. For any character φ , let $u_\varphi = \sum_{\delta \in \Delta} \varphi(\delta) \delta^{-1}$. Then, we have

$$\forall \varphi \in \Delta^*, u_\varphi u_\psi = \begin{cases} |\Delta| u_\varphi & \text{if } \varphi = \psi, \\ 0 & \text{else.} \end{cases}$$

Moreover, we have

$$\sum_{\varphi \in \Delta^*} u_\varphi = |\Delta|.$$

Then, one notices that the multiplication-by-2 maps $\Lambda[\Delta]$ to the sum of the $u_\varphi \Lambda[\Delta]$'s. So, we get a pseudo-isomorphism with cokernel a group of exponent 2 when $p = 2$.

Finally, we need to study $\text{Hom}_G(\Lambda[\Delta], C)$ and $\text{Hom}_G(\mathbf{Z}_p(1), C)$ to conclude. For the first one, notice that a continuous G -module morphism from $\Lambda[\Delta]$ to C is uniquely defined by its image of a topological generator of H . Hence, $\text{Hom}_G(\Lambda[\Delta], C)$ is isomorphic to $\mathbf{Q}_p/\mathbf{Z}_p$. Then, let φ be a continuous G -module morphism from $\mathbf{Z}_p(1)$ to C . By continuity, φ is uniquely determined by $\varphi(1)$. Moreover, we have:

$$\forall g \in G, \psi(g)\varphi(1) = \chi(g)\varphi(1).$$

By hypothesis, ψ is different from χ . Hence, we can find $g_0 \in G$ such that $a = (\psi(g_0) - \chi(g_0))$ is non-zero and annihilates $\varphi(1)$. In particular, if v is the p -adic valuation of a , then the kernel of a is given by elements of order at most p^v . In other words, $\varphi(1)$ belongs to $\mathbf{Q}_p/\mathbf{Z}_p[p^v] \cong \mathbf{Z}/p^v\mathbf{Z}$. Therefore, $\text{Hom}_G(\mathbf{Z}_p(1), C)$ is finite.

□

Let us denote by ψ and ϕ the actions of G_v on C_v and $\tilde{E}[p^\infty]$ respectively. Since ψ and ϕ induce representations of $C_v[p^n]$ and $\tilde{E}[p^n]$ for any integer n , they induce representations of the Tate-modules $T_p(C_v) = \varprojlim_n C_v[p^n]$ and $T_p(\tilde{E})$. In particular, in a suitable basis, the action of G_v on $T_p(E)$ is given by

$$\begin{pmatrix} \psi & * \\ 0 & \phi \end{pmatrix}$$

since C_v is stable under the action of G_v . By the Weil-pairing (see Section 3.1), the action of G_v on $T_p(E) \wedge T_p(E)$ is given by the determinant of the above matrix while its action on $\mathbf{Z}_p(1)$ is given by the cyclotomic character χ . Therefore, we have $\psi\phi = \chi$.

Suppose ϕ^m is the trivial action for some positive integer m . In particular, applying ϕ^m to the Frobenius map, we get

$$\forall (x, y) \in \tilde{E}[p^\infty], (x^{q^m}, y^{q^m}) = (x, y).$$

where q is the cardinal of f_v . However, $X^{q^m} - X$ has a finite number of roots and $\tilde{E}[p^\infty]$ is infinite (remember it is isomorphic to $\mathbf{Q}_p/\mathbf{Z}_p$). By contradiction, ϕ has infinite order.

Let I_v be the inertia subgroup of G_v . By definition, $\phi|_{I_v}$ is trivial so we have $\psi|_{I_v} = \chi|_{I_v}$. Since I_v is non-trivial, this means that ψ has infinite order. In particular, ψ is non-trivial. Furthermore, $\phi|_{G_\eta}$ is non-trivial. Indeed, the Frobenius map of $\tilde{E}(m_\eta)$ is the $[m_\eta : f_v]$ -power of the Frobenius map of $\tilde{E}$. Hence, $\phi|_{G_\eta}$ has infinite order. So $\psi|_{G_\eta}$ is not the cyclotomic character. Thus, the proposition follows from Lemma 3.2.9. $\square$

Proposition 3.2.10. *Suppose w divides p and let M_η be a finite extension of F_w . Then, we have*

$$\frac{\text{Im}(\lambda_\eta)}{\text{Im}(\kappa_\eta)} \cong \tilde{E}(m_\eta)_p$$

where m_η denotes the residue field of M_η .

Proof. Let us denote by $\mathcal{F}$ the formal group for E . Let $\mathfrak{m}$ denote the maximal ideal of M_η . The p -adic logarithm induces a group isomorphism between $\mathcal{F}(\overline{\mathfrak{m}})$ and $\overline{\mathfrak{m}}$. Hence, $\mathcal{F}(\overline{\mathfrak{m}})$ is divisible and we can build a Kummer sequence and define $\kappa_{\mathcal{F}}$ analogously as κ_η (one needs to consider the exact sequence associated to the multiplication-by- p^m map on $\mathcal{F}(\mathfrak{m})$ that is surjective). So, we build the following commutative diagram whose rows are exact:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}(\mathfrak{m}) \otimes (\mathbf{Q}_p/\mathbf{Z}_p) & \xrightarrow{\kappa_{\mathcal{F}}} & H^1(M_\eta, C_w) & \longrightarrow & H^1(M_\eta, \mathcal{F}(\overline{\mathfrak{m}}))_p \longrightarrow 0 \\ & & \downarrow & & \downarrow \lambda_\eta & & \downarrow \epsilon \\ 0 & \longrightarrow & E(M_\eta) \otimes (\mathbf{Q}_p/\mathbf{Z}_p) & \xrightarrow{\kappa_\eta} & H^1(M_\eta, E[p^\infty]) & \longrightarrow & H^1(M_\eta, E(\overline{F_w}))_p \longrightarrow 0 \end{array} \quad (3.6)$$

Now, consider the exact sequence where m_η is the residue field of M_η :

$$0 \longrightarrow \mathcal{F}(\mathfrak{m}) \longrightarrow E(M_\eta) \longrightarrow \tilde{E}(m_\eta) \longrightarrow 0$$

By Proposition 2.1.3, we have

$$H^1(M_\eta, C_w) = \varinjlim_n H^1(M_\eta, \mathbf{Z}/p^n \mathbf{Z})$$

so $H^1(M_\eta, C_w)$ is a p -primary group as $\mathcal{F}(\mathfrak{m}) \otimes \mathbf{Q}_p/\mathbf{Z}_p$. According to Lemma 3.2.9, the $\mathbf{Z}_p$ -corank of $H^1(M_\eta, C_w)$ is $[M_\eta : \mathbf{Q}_p]$. Moreover, $\mathcal{F}(\mathfrak{m})$ is isomorphic to $\mathfrak{m}$, itself isomorphic to $\mathbf{Z}_p^{[M_\eta : \mathbf{Q}_p]}$. Since $\mathbf{Z}_p \otimes (\mathbf{Q}_p/\mathbf{Z}_p)$ is isomorphic to $\mathbf{Q}_p/\mathbf{Z}_p$,

we conclude that the $\mathbf{Z}_p$ -corank of $\mathcal{F}(\mathfrak{m}) \otimes (\mathbf{Q}_p/\mathbf{Z}_p)$ and $H^1(M_\eta, C_w)$ coincide. Moreover, since $\mathcal{F}(\mathfrak{m}) \otimes (\mathbf{Q}_p/\mathbf{Z}_p)$ is divisible, we deduce that $\text{Im}(\kappa_{\mathcal{F}})$ is equal to $H^1(M_\eta, C_w)_{\text{div}}$. Hence, (3.6) and Proposition 3.2.8 yield the following commutative diagram whose rows are exact

$$\begin{array}{ccccccc} 0 & \longrightarrow & \frac{H^1(M_\eta, C_w)}{H^1(M_\eta, C_w)_{\text{div}}} & \longrightarrow & H^1(M_\eta, \mathcal{F}(\overline{\mathfrak{m}}))_p & \longrightarrow & 0 \\ & & \downarrow \theta & & \downarrow \epsilon & & \\ 0 & \longrightarrow & \frac{\text{Im}(\lambda_\eta)}{\text{Im}(\lambda_\eta)_{\text{div}}} & \longrightarrow & H^1(M_\eta, E(\overline{F_w}))_p & & \end{array}$$

The map θ is surjective by construction, and its kernel is isomorphic to $\ker(\epsilon)$ according to the diagram above. Moreover, from the short exact sequence

$$0 \longrightarrow \mathcal{F}(\overline{\mathfrak{m}}) \longrightarrow E(\overline{F_w}) \longrightarrow \tilde{E}(\overline{f_w}) \longrightarrow 0$$

where f_w denotes the residue field of F_w , we get by Galois cohomology long exact sequence the following exact sequence:

$$E(M_\eta) \longrightarrow \tilde{E}(m_\eta) \longrightarrow H^1(M_\eta, \mathcal{F}(\overline{\mathfrak{m}})) \xrightarrow{\epsilon'} H^1(M_\eta, E(\overline{F_w}))$$

By surjectivity of the first morphism, we deduce that ϵ' is injective. Hence, since, ϵ is induced by ϵ' , it inherits its injectivity. Therefore, θ is an isomorphism. Hence, we are reduced to studying $H^1(M_\eta, C_w)/H^1(M_\eta, C_w)_{\text{div}}$.

Let us consider C as in the proof of Lemma 3.2.9, i.e., $C = \mathbf{Q}_p/\mathbf{Z}_p(\psi)$. We denote by T the Tate-module of C . In other words, T is the inverse limit of the $C[p^n]$'s, equipped with the action induced by ψ . Let $V = T \otimes_{\mathbf{Z}_p} \mathbf{Q}_p$. Then, we have the following exact sequence:

$$0 \longrightarrow T \longrightarrow V \longrightarrow C \longrightarrow 0$$

By Galois cohomology long exact sequence, we get:

$$H^1(M_\eta, V) \xrightarrow{\alpha} H^1(M_\eta, C) \xrightarrow{\beta} H^2(M_\eta, T) \xrightarrow{\gamma} H^2(M_\eta, V)$$

Since $H^1(M_\eta, C)$ is torsion, $\text{Im}(\beta)$ is included in $H^2(M_\eta, T)_{\text{tors}}$. Moreover, $H^2(M_\eta, V)$ inherits a structure of $\mathbf{Q}_p$ -vector space since V is a $\mathbf{Q}_p$ -vector space. Hence, $H^2(M_\eta, V)$ has no torsion and $H^2(M_\eta, C)_{\text{tors}}$ is included in the kernel of γ . Therefore, $\ker(\gamma)$ is equal to $H^2(M_\eta, T)_{\text{tors}}$.

By Theorem 2.1.5, $H^2(M_\eta, T)$ is isomorphic to $H^0(M_\eta, \text{Hom}(T, \mu_{p^\infty}))$, i.e., to $\text{Hom}_{G_\eta}(T, \mu_{p^\infty})$. Since G_η acts on T by ψ and on μ_{p^∞} by χ , $\text{Hom}_{G_\eta}(T, \mu_{p^\infty})$ can be identified with $H^0(M_\eta, (\mathbf{Q}_p/\mathbf{Z}_p)(\chi\psi^{-1}))$. Hence, we have

$$\text{coker}(\alpha) \cong \ker(\gamma) \cong H^0(M_\eta, (\mathbf{Q}_p/\mathbf{Z}_p)(\chi\psi^{-1})).$$

Denoting by ϕ the action of G_η on $\tilde{E}[p^\infty]$ as in the proof of Proposition 3.2.8, we have $\chi\psi^{-1} = \phi$. Hence, $H^0(M_\eta, (\mathbf{Q}_p/\mathbf{Z}_p)(\chi\psi^{-1}))$ is isomorphic to $\tilde{E}(m_\eta)_p$.

Now, notice that $H^1(M_\eta, V)$ is a $\mathbf{Q}_p$ -vector space and thereby is divisible. So, $\text{Im}(\alpha)$ is divisible, and since $\text{coker}(\alpha)$ is finite, it does not contain any non trivial divisible subgroup. In other words, $\text{Im}(\alpha)$ is equal to $H^1(M_\eta, C)_{\text{div}}$. Thus, $H^1(M_\eta, C)/H^1(M_\eta, C)_{\text{div}}$ is isomorphic to $\tilde{E}(m_\eta)_p$. $\square$

Proposition 3.2.11. *Let K/F_w be a Galois extension. Assume $\text{Gal}(K/F_w)$ contains an infinite pro- p subgroup and that its inertia subgroup is of finite index. Then, $\text{Im}(\kappa_K) = \text{Im}(\lambda_K)$ where κ_K is the Kummer morphism for E over K and λ_K is the canonical morphism from $H^1(K, C_w)$ to $H^1(K, E[p^\infty])$.*

Proof. Let K as above. The images of κ_K and λ_K are the direct limits of the images of $\kappa_{K'}$ and $\lambda_{K'}$ where K' runs through finite subextensions of K/F_w . Let us fix such a K' . According to Proposition 3.2.8, we have $\text{Im}(\kappa_{K'}) = \text{Im}(\lambda_{K'})_{\text{div}}$. Besides, according to Proposition 3.2.10, $\text{Im}(\lambda_{K'})/\text{Im}(\lambda_{K'})_{\text{div}}$ has order bounded by $|\tilde{E}(k)_p|$ where k denotes the residue field of K . Note that the cardinal of $\tilde{E}(k)$ is finite because k is finite since we assume the inertia subgroup of $\text{Gal}(K/F_w)$ of finite index. So, we conclude that $\text{Im}(\lambda_{K'})/\text{Im}(\kappa_{K'})$ is finite. Then, since C_w is divisible, the following sequence is exact

$$0 \longrightarrow C_w[m] \longrightarrow C_w \xrightarrow{m} C_w \longrightarrow 0$$

By Galois cohomology long exact sequence, we get

$$H^1(K, C_w) \xrightarrow{m} H^1(K, C_w) \longrightarrow H^2(K, C_w[m])$$

Since C_w is p -primary, if p does not divide m then the multiplication-by- m map is an isomorphism so $H^2(K, C_w[m])$ is trivial. Moreover, according to Serre [10, Chapter II, §3], $\text{Gal}(\overline{K}/K)$ has p -cohomological dimension 1 since it contains an infinite pro- p group, i.e., $H^s(K, N)$ is trivial for any integer s greater than 1 and any p -primary $\text{Gal}(\overline{K}/K)$ -module N . Hence, $H^2(K, C_w[p])$ is trivial. Therefore, $H^1(K, C_w)$ is divisible. The image of a divisible group is still divisible, so $\text{Im}(\lambda_K)$ is divisible and so is the quotient $\text{Im}(\lambda_K)/\text{Im}(\kappa_K)$. A finite divisible group is always trivial so we get the result. $\square$

3.3 Control theorem

Let M be an algebraic extension of F , p be a prime number and m be an integer. If η is a prime of M , we denote by $\mathcal{H}_E(M_\eta)$ the quotient $H^1(M_\eta, E[p^\infty])/\text{Im}(\kappa_\eta)$. Moreover, let $\mathcal{P}(E) = \prod_{\eta \text{ prime}} \mathcal{H}_E(M_\eta)$, so $\text{Sel}_E(M)_p$ is the kernel of the map from $H^1(M, E[p^\infty])$ to $\mathcal{P}(E)$ induced by restrictions to decomposition groups. We denote by $\mathcal{G}_E(M)$ the image of this last map.

Let us consider the $\mathbf{Z}_p$ -extension F_∞/F , i.e., a Galois extension such that $\text{Gal}(F_\infty/F) \cong \mathbf{Z}_p$ and F_∞ is a subfield of $F(\mu_{p^\infty})$. Thanks to Galois theory, we can build a tower of extensions $F = F_0 \subset F_1 \subset F_2 \subset \dots$ such that $\text{Gal}(F_m/F) \cong \mathbf{Z}/p^m\mathbf{Z}$ and $F_\infty = \bigcup_{n \in \mathbf{N}} F_n$. We denote by Γ_m the Galois group $\text{Gal}(F_\infty/F_m)$. By Galois theory, this is the p^m -th power of Γ_0 , denoted Γ .

Now, consider the following commutative diagram whose rows are exact:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Sel}_E(F_m)_p & \longrightarrow & H^1(F_m, E[p^\infty]) & \longrightarrow & \mathcal{G}_E(F_m) \longrightarrow 0 \\
& & \downarrow s_m & & \downarrow h_m & & \downarrow g_m \\
0 & \longrightarrow & \mathrm{Sel}_E(F_\infty)_p^{\Gamma_m} & \longrightarrow & H^1(F_\infty, E[p^\infty])^{\Gamma_m} & \longrightarrow & \mathcal{G}_E(F_\infty)^{\Gamma_m}
\end{array}$$

The morphisms s_m , h_m and g_m are induced by inflation-restriction sequences.

Our goal is to show that s_n is a pseudo-isomorphism, i.e., its kernel and cokernel are finite. By the snake's lemma (see Lang [16, Chapter III, Lemma 9.1] for instance), we have the exact sequence

$$0 \rightarrow \ker(s_m) \rightarrow \ker(h_m) \rightarrow \ker(g_m) \rightarrow \mathrm{coker}(s_m) \rightarrow \mathrm{coker}(h_m)$$

Hence, we need to study the cokernel of h_m and the kernels of h_m and g_m .

Lemma 3.3.1. *The kernel of h_n is finite of bounded order as n varies through the natural integers.*

Proof. Let us denote by B the p -primary subgroup of $E(F_\infty)$. From the inflation-restriction sequence, we get the following exact sequence:

$$0 \longrightarrow H^1(\Gamma_m, B) \longrightarrow H^1(F_m, E[p^\infty]) \xrightarrow{h_m} H^1(F_\infty, E[p^\infty])^{\Gamma_m}$$

noticing that $\Gamma_m \cong \mathrm{Gal}(\overline{F}/F_m)/\mathrm{Gal}(\overline{F}/F_\infty)$ by Galois theory. Hence, the kernel of h_m is isomorphic to $H^1(\Gamma_m, B)$. Let γ be a topological generator of Γ . Then, γ^{p^m} is a topological generator of Γ_m , and by Proposition 2.3.2, $H^1(\Gamma_m, B)$ is isomorphic to $B/(\gamma^{p^m} - 1)B$. By the Mordell-Weil theorem (see Silverman [15, Chapter VIII, Theorem 6.7]), $E(F_m)$ is finitely generated. Hence, since the kernel of $\gamma^{p^m} - 1$ is $E(F_m)[p^\infty]$ (the set of elements of B fixed by γ^{p^m} and thereby by Γ_m), it is finite.

Now, notice that $E(F_m)[p^\infty]$ is contained in $E(\overline{F})[p^\infty]$ which is isomorphic to $(\mathbf{Q}_p/\mathbf{Z}_p)^2$. Hence, B_{div} has finite $\mathbf{Z}_p$ -corank, at most 2 according to Corollary 2.2.7. In particular, since the kernel of $\gamma^{p^m} - 1$ is finite, its image has $\mathbf{Z}_p$ -corank equal to $\mathrm{corank}_{\mathbf{Z}_p}(B)$. This implies:

$$B_{\mathrm{div}} \subset (\gamma^{p^m} - 1)B \subset B$$

From the above inclusion, we deduce that the order of $H^1(\Gamma_m, B)$ is bounded by $[B : B_{\mathrm{div}}]$, which is finite. $\square$

Lemma 3.3.2. *The cokernel of h_n is trivial for any integer n .*

Proof. Let us denote by B the p -primary subgroup of $E(F_\infty)$. From the inflation-restriction sequence, we get the following exact sequence:

$$H^1(F_m, E[p^\infty]) \xrightarrow{h_m} H^1(F_\infty, E[p^\infty])^{\Gamma_m} \longrightarrow H^2(\Gamma_m, B)$$

However, Γ_m is a free pro- p group hence $H^2(\Gamma_m, B) = 0$ (see Section 2.3). Thus, h_m is surjective. $\square$

For any prime v of F , we denote by v_m a prime of F_m lying above v . Let w be a prime of F . Let us introduce

$$r_{w_m} : \mathcal{H}_E((F_m)_{w_m}) \longrightarrow \mathcal{H}_E((F_\infty)_\eta)$$

where η is any prime of F_∞ lying above w_m . We are going to study the kernel of the map r_{w_m} that does not depend on the choice of η .

Since we consider a $\mathbf{Z}_p$ -cyclotomic extension, $F_w = F_\eta$ whenever w is archimedean according to Washington [14, Proposition 13.2]. Consequently, if w is archimedean then $\ker_{w_m}$ is trivial. From now on, we assume that w is non-archimedean.

Lemma 3.3.3. *Suppose that w does not divide p . Then $\ker(r_{w_n})$ is finite of bounded order as n varies through $\mathbf{N}$. Moreover, if E has good reduction at p then $\ker(r_{w_n})$ is trivial for any integer n .*

Proof. According to Proposition 3.2.7, $\mathcal{H}_E(M_\eta) = H^1(M_\eta, E[p^\infty])$ for any algebraic extension M_η/F_w . Let $K = (F_\infty)_\eta$ and $B_w = H^0(K, E[p^\infty])$. Since w does not divide p , it is unramified and finitely decomposed in F_∞/F (see Neukirch [17, Chapter I, Proposition 10.3] for details). Hence, there is a natural integer s such that for $N \in \mathbf{N}$ big enough, there are exactly s primes above w in F_N for any natural integer $N' \geq N$. In particular, for any natural integer $N' \geq N$, since the extension $F_{N'}/F_N$ is Galois and of Galois group cyclic of order a power of p , the extension $(F_{N'})_{v'}/(F_N)_v$ has a Galois group cyclic of order a power of p for any primes v' and v such that $v'|v|w$. Hence, K is the unramified $\mathbf{Z}_p$ -extension of F_w . As in the proof of Lemma 3.3.1, B_w is isomorphic to $(\mathbf{Q}_p/\mathbf{Z}_p)^e \times \Delta$ where $e \in \{0, 1, 2\}$ and Δ is a finite group. Let $\Gamma_{w_m} = \text{Gal}(K/(F_m)_{w_m})$. This group is isomorphic to $\mathbf{Z}_p$. Let γ_{w_m} be a topological generator of Γ_{w_m} . Then, we deduce from the inflation-restriction sequence

$$0 \longrightarrow H^1(\Gamma_{w_m}, B_w) \longrightarrow H^1((F_m)_{w_m}, E[p^\infty]) \xrightarrow{r_{w_m}} H^1(K, E[p^\infty])^{\Gamma_{w_m}}$$

that

$$\ker(r_{w_m}) \cong H^1(\Gamma_{w_m}, B_w) \cong \frac{B_w}{(\gamma^{p^m} - 1)B_w}.$$

As in the proof of Lemma 3.3.1, $(\gamma^{p^m} - 1)B_w$ contains $(B_w)_{\text{div}}$ since $E((F_m)_w)$ has a finite p -primary subgroup. Indeed, this last assertion has been proven in Proposition 3.2.7. We deduce

$$|\ker(r_{w_m})| \leq [B_w : (B_w)_{\text{div}}]$$

where the right-hand side is finite and independent of m . If m is big enough, this equality becomes an equality.

Suppose E has good reduction at w . Let us denote by L the field F_w extended with all coordinates of points of $E[p^\infty]$. Then, since w does not divide p the field extension L/F_w is unramified thanks to the criterion of Néron-Ogg-Shafarevich

(see Silverman [15, Chapter VII, Theorem 7.1]). Let us consider the Weil pairing as in Section 3.1:

$$e: E[p^\infty] \times E[p^\infty] \longrightarrow \mu_{p^\infty}.$$

By definition, $E[p^\infty]$ is included in $E(L)$. Hence, according to Silverman [15, Chapter III, Corollary 8.1.1], L contains p -power roots of unity. Since K is a $\mathbf{Z}_p$ -cyclotomic extension, we deduce that it is contained in L . Now, let $\Delta = \text{Gal}(L/K)$. Since L/K is unramified, Δ embeds into $\text{Gal}(l/k)$ where l and k denote the residue fields of L and K respectively. In particular, Δ is a quotient of a subgroup of $\text{Gal}(\bar{f}/f)$ where f denotes the residue field of F_w . This Galois group is the projective limit of the $\text{Gal}(f'/f)$ where f' runs through finite extension of f . Then, for any finite extension f'/f , $\text{Gal}(f'/f)$ is a cyclic group. Hence, $\text{Gal}(\bar{f}/f)$ is isomorphic to $\hat{\mathbf{Z}}$. Thus, since K is a $\mathbf{Z}_p$ -extension, Δ is a quotient of $\prod_{q \neq p} \mathbf{Z}_q$. Besides, Δ embeds into $\mathcal{GL}_2(\mathbf{Z}_p)$ by definition of L . Consider the reduction map $\rho: \mathcal{GL}_2(\mathbf{Z}_p) \rightarrow \mathcal{GL}_2(\mathbf{Z}/p\mathbf{Z})$. Then, the kernel of ρ is isomorphic to $1 + p\mathcal{M}_2(\mathbf{Z}_p)$ and in particular, is isomorphic to $\mathbf{Z}_p^4$. In other words, $\ker(\rho)$ is a pro- p group. Then, the image of Δ into $\mathcal{GL}_2(\mathbf{Z}_p)$ intersected with $\ker(\rho)$ should be trivial. Hence, Δ embeds into $\mathcal{GL}_2(\mathbf{Z}/p\mathbf{Z})$ and is thereby finite. Since it is finite, it is also cyclic of order prime with p as a quotient of $\prod_{q \neq p} \mathbf{Z}_q$. For more details on quotients of $\hat{\mathbf{Z}}$, we refer the reader to the book of Neukirch, Schmidt and Wingberg [12, Chapter I, §7].

Now, notice that B_w is equal to $E[p^\infty]^\Delta$. Hence, it is sufficient to show that $E[p^\infty]^\Delta$ is divisible to conclude. Let P be a point from $E[p^\infty]^\Delta$. Since $E[p^\infty]$ is divisible, there is a point $Q \in E[p^\infty]$ such that $mQ = P$. Since the order of Δ is prime with p , $|\Delta|$ is invertible. Hence, we can write

$$Q' = \frac{1}{|\Delta|} \sum_{\sigma \in \Delta} Q^\sigma.$$

The point Q' is clearly a point from $E[p^\infty]^\Delta$. Besides, we have $mQ' = P$ so $E[p^\infty]^\Delta$ is divisible. $\square$

Now, let us assume that w divides p . For any integer n , let f_{w_n} denote the residue field of $(F_n)_{w_n}$. The decomposition group G_η has finite index in $\text{Gal}(F_\infty/F)$ so for n big enough, $\text{Gal}(F_\infty/F_n)$ is included in G_η and thereby, w_n is totally ramified in F_∞/F . Thus, f_{w_n} is eventually equal to f_η , the residue field of $(F_\infty)_\eta$. We denote by $\tilde{E}$ the reduction of E at w .

Lemma 3.3.4. *Assume that E has good, ordinary reduction at w . Then*

$$|\ker(r_{w_n})| = |\tilde{E}(f_{w_n})_p|^2$$

is finite and bounded as n varies through $\mathbf{N}$.

Proof. Assume that E has good, ordinary reduction at w . Let C_w denote the kernel of the reduction map from $E[p^\infty]$ to $\tilde{E}[p^\infty]$. According to Proposition 3.2.8, we have $\text{Im}(\kappa_{w_m}) = \text{Im}(\lambda_{w_m})$. The inertia subgroup of $\text{Gal}(F_\infty/F_n)$ for w_n is isomorphic to $\text{Gal}(F_\infty/F_n)$ for n big enough according to the discussion above.

In particular, since a subgroup of $\mathbf{Z}_p$ isomorphic to $\mathbf{Z}_p$ is of the form $p^s \mathbf{Z}_p$ for some integer s , the inertia subgroup of $\text{Gal}(F_\infty/F)$ for w has finite index. So, $\text{Im}(\kappa_\eta) = \text{Im}(\lambda_\eta)$ according to Proposition 3.2.11. Therefore, r_{w_m} factors as follows:

$$\begin{array}{ccc} \frac{H^1((F_m)_{w_m}, E[p^\infty])}{\text{Im}(\lambda_{w_m})_{\text{div}}} & \xrightarrow{a_{w_m}} & \frac{H^1((F_m)_{w_m}, E[p^\infty])}{\text{Im}(\lambda_{w_m})} \\ & \searrow r_{w_m} & \downarrow b_{w_m} \\ & & \frac{H^1((F_\infty)_\eta, E[p^\infty])}{\text{Im}(\lambda_\eta)} \end{array}$$

□

The map a_{w_m} is surjective since $\text{Im}(\lambda_{w_m})_{\text{div}}$ is included in $\text{Im}(\lambda_{w_m})$. So, we have

$$|\ker(r_{w_m})| = |\ker(a_{w_m})| |\ker(b_{w_m})|.$$

In addition, according to Proposition 3.2.10, we have $|\ker(a_{w_m})| = |\tilde{E}(f_{w_m})_p|$. For what concerns the kernel of b_{w_m} , we consider the following commutative diagram whose rows are exact, given by the Galois cohomology long sequence:

$$\begin{array}{ccccccc} H^1((F_m)_{w_m}, C_w) & \xrightarrow{\lambda_{w_m}} & H^1((F_m)_{w_m}, E[p^\infty]) & \xrightarrow{\pi_{w_m}} & H^1((F_m)_{w_m}, \tilde{E}[p^\infty]) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow d_{w_m} & & \\ H^1((F_\infty)_\eta, C_w) & \xrightarrow{\lambda_\eta} & H^1((F_\infty)_\eta, E[p^\infty]) & \xrightarrow{\pi_\eta} & H^1((F_\infty)_\eta, \tilde{E}[p^\infty]) & \longrightarrow & 0 \end{array}$$

By Theorem 2.1.5, we have $H^2(M, C_w) = H^0(M, C_w(-1))$ for any finite extension $M/(F_m)_{w_m}$, where $C_w(-1)$ denotes C_w seen as a $\text{Gal}(\bar{M}/M)$ -module with an action given by the inverse of the cyclotomic character, i.e., by χ^{-1} . By Kummer theory, in particular thanks to the discussion in Lemma 3.2.9, we have $H^2(M, C_w) = 0$ for any such field M . Hence, this gives the surjectivity of π_{w_m} . Since $\text{Gal}(\bar{f}_w/f_{w_n})$ is equal to $\text{Gal}(\bar{f}_w/f_\eta)$ for n big enough, we deduce also the surjectivity of π_η .

By exactness and commutativity of the above diagram, $\ker(b_{w_m})$ is isomorphic to $\ker(d_{w_m})$. Now, by the inflation-restriction sequence, we have

$$\ker(d_{w_m}) \cong H^1(\text{Gal}((F_\infty)_\eta/(F_m)_{w_m}), \tilde{E}(f_\eta)_p) \cong \frac{\tilde{E}(f_\eta)_p}{(\gamma_{w_m} - 1)\tilde{E}(f_\eta)_p}$$

where γ_{w_m} is a topological generator of $\text{Gal}((F_\infty)_\eta/(F_m)_{w_m})$ and the second isomorphism is given by Proposition 2.3.2. Since $\gamma_{w_m} - 1$ is an endomorphism, its kernel and cokernel have same order. Since its kernel is $\tilde{E}(f_{w_m})_p$, we deduce that the order of $\ker(d_{w_m})$ is $|\tilde{E}(f_{w_m})_p|$.

Lemma 3.3.5. *The kernel of g_n is finite of bounded order as n varies through $\mathbf{N}$.*

Proof. Let Σ_0 denote the finite set of nonarchimedean primes of F which either lie above p or where E has bad reduction. According to Lemma 3.3.3 and the discussion about archimedean places, if v is a place not belonging to Σ_0 and v_m is a place of F_m lying above v then $\ker(r_{v_m}) = 0$. Moreover, Lemma 3.3.3 and Lemma 3.3.4 show that $\ker(r_{v_n})$ is finite of bounded order as n varies through $\mathbf{N}$ for any $v \in \Sigma_0$. In addition, the number of primes of F_n lying above a nonarchimedean place v is bounded as n varies through $\mathbf{N}$. $\square$

Thanks to the study of h_m and g_m , we conclude that s_m is a pseudo-isomorphism. This is the content of the celebrated Mazur's control theorem:

Theorem 3.3.6 (Mazur's control theorem). *Assume that E has good, ordinary reduction at all primes of F lying above p . Then, the maps*

$$s_n : \mathrm{Sel}_E(F_n)_p \longrightarrow \mathrm{Sel}_E(F_\infty)_p^{\Gamma_n}$$

have finite kernel and cokernel, of bounded order as n varies through $\mathbf{N}$.

Manin [27, 28] suggested that the reduction behavior hypothesis on E could be reduced as illustrated by the following conjecture.

Conjecture 3.3.7. *Assume that E has good, ordinary or multiplicative reduction at all primes of F lying above p . Then, the maps*

$$s_n : \mathrm{Sel}_E(F_n)_p \longrightarrow \mathrm{Sel}_E(F_\infty)_p^{\Gamma_n}$$

have finite kernel and cokernel, of bounded order as n varies through $\mathbf{N}$.

A result of Barre, Sirieix, Diaz, Gramain and Philibert [29] implies that this conjecture holds when $F = \mathbf{Q}$.

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